MODULE 2- CONTROL FLOW

- **if-else**
- **switch**
- **Loops**
- **break / continue**

Control Flow in JavaScript

Control Flow determines the order in which statements are executed in a program.

Normally, JavaScript executes code from **top to bottom**.

```
console.log("Step 1");
console.log("Step 2");
console.log("Step 3");
```

Output

```
Step 1
Step 2
Step 3
```

Sometimes we want to execute code only when a condition is true. For this, JavaScript provides **Control Flow Statements**.

What is an if Statement?

The if statement executes a block of code only when a specified condition is **true**.

Syntax

```
if(condition)
{
   // code to execute
}
```

Types of if Statements

1. Simple if
2. if...else
3. else if Ladder
4. Nested if
5. switch Statement (Alternative to multiple conditions)

---

1. Simple if Statement

Executes code only when the condition is true.

Program

```
<!DOCTYPE html>
<html>
<body>
<script>
let age = 20;
if(age >= 18)
{
   document.write("You can vote");
}
</script>
```

```
</body>
</html>
```

Output

You can vote

Example: Positive Number

```
<!DOCTYPE html>
<html>
<body>

<script>

let num = 15;

if(num > 0)
{
   document.write("Positive Number");
}

</script>

</body>
</html>
```

Output

Positive Number

---

2. if...else Statement

Executes one block when the condition is true and another block when it is false.

Syntax

```
if(condition)
{
   // true block
}
else
{
   // false block
}
```

Program: Even/Odd

```
<!DOCTYPE html>
<html>
<body>

<script>

let num = 7;

if(num % 2 == 0)
{
   document.write("Even Number");
}
else
{
   document.write("Odd Number");
}
```

```
</script>
</body>
</html>
```

Output

Odd Number

Program: Voting Eligibility

```
<!DOCTYPE html>
<html>
<body>
<script>
let age = 16;
if(age >= 18)
{
   document.write("Eligible for Voting");
}
else
{
   document.write("Not Eligible");
}

</script>
</body>
</html>
```

Output

Not Eligible

---

3. else if Ladder

Used when there are multiple conditions.

Syntax

```
if(condition1)
{
}
else if(condition2)
{
}
else if(condition3)
{
}
else
{
}
```

Program: Grade Calculator

```
<!DOCTYPE html>
<html>
<body>

<script>
let marks = 75;

if(marks >= 90)
{
   document.write("Grade A+");
}
```

```
else if(marks >= 80)
{
   document.write("Grade A");
}
else if(marks >= 70)
{
   document.write("Grade B");
}
else if(marks >= 60)
{
   document.write("Grade C");
}
else
{
   document.write("Fail");
}

</script>

</body>
</html>
```

Output

Grade B

---

4. Nested if Statement

An if inside another if.

Syntax

```
if(condition1)
{
   if(condition2)
   {
   }
}
```

Program: Login Verification

```
<!DOCTYPE html>
<html>
<body>

<script>

let username = "admin";
let password = "1234";

if(username == "admin")
{
   if(password == "1234")
   {
      document.write("Login Successful");
   }
   else
   {
      document.write("Wrong Password");
   }
}
else
{
```

```
    document.write("Invalid Username");
}

</script>

</body>
</html>
```

Output

Login Successful

---

5. switch Statement

Used when checking multiple values of the same variable.

Syntax

```
switch(expression)
{
   case value1:
      statement;
      break;

   case value2:
      statement;
      break;

   default:
      statement;
}
```

Program: Day Name

```
<!DOCTYPE html>
<html>
<body>

<script>

let day = 3;

switch(day)
{
   case 1:
      document.write("Monday");
      break;

   case 2:
      document.write("Tuesday");
      break;

   case 3:
      document.write("Wednesday");
      break;

   case 4:
      document.write("Thursday");
      break;

   default:
      document.write("Invalid Day");
}
```

```
</script>

</body>
</html>
```

Output

Wednesday

Ques : Check Number is Even / ODD ?

Method 1: Using Bitwise AND (&) ?

```
let num = 10;

console.log(num & 1 ? "Odd" : "Even");
```

Output

Even

How it works?

Even Number → Last bit = 0
Odd Number → Last bit = 1

Examples:

```
console.log(8 & 1); // 0
console.log(9 & 1); // 1
```

This is a very fast method and is often asked in interviews.

Method 2: One-Line Arrow Function

```
const check = n => n % 2 ? "Odd" : "Even";
const mysum=(a,b,c)=>a+b+c;

console.log(check(15));
console.log(mysum(1,2,3))
```

Output

Odd

Notice that n % 2 itself returns:

- 0 for even (false)
- 1 for odd (true)

So no need to write == 0.

Method 3: Using Array Indexing (Fun Trick)

```
let num = 7;
let a=["Even", "Odd"];
console.log(a[num % 2]);
```

Output

Odd

Explanation:

num % 2 = 0 → Array[0] = "Even"
num % 2 = 1 → Array[1] = "Odd"

Method 4: Function Shortcut

```
function evenOdd(n) {
   return n % 2 ? "Odd" : "Even";
}

console.log(evenOdd(20));
```

Output

Even

Best Method for Teaching Students

```
num % 2 == 0 ? "Even" : "Odd"
```

Best Method for Interviews

```
num & 1 ? "Odd" : "Even"
```

Shortest Readable Method

```
num % 2 ? "Odd" : "Even"
```

because:

0 → false → Even
1 → true → Odd

Real-World Example: Student Result

```
<!DOCTYPE html>
<html>
<body>

<script>

let marks = 82;

if(marks >= 33)
{
   document.write("Pass");
```

```
}
else
{
   document.write("Fail");
}

</script>

</body>
</html>
```

Menu Driven Program Using switch

```
<!DOCTYPE html>
<html>
<body>

<script>

let choice = 2;
switch(choice)
{
   case 1:
      document.write("Add Record");
      break;
   case 2:
      document.write("Update Record");
      break;
   case 3:
      document.write("Delete Record");
      break;
```

```
    default:
        document.write("Invalid Choice");
}
</script>
</body>
</html>
```

Output

Update Record

Comparison Table

Statement	Purpose
if	One condition
if...else	Two possible outcomes
else if	Multiple conditions
Nested if	Condition inside condition
switch	Multiple fixed values

Complete Demonstration Program

```
<!DOCTYPE html>
<html>
<head>
<title>Control Flow Example</title>
</head>
<body>

<script>
```

```
let marks = 85;

if(marks >= 90)
{
   document.write("Grade A+");
}
else if(marks >= 80)
{
   document.write("Grade A");
}
else if(marks >= 70)
{
   document.write("Grade B");
}
else
{
   document.write("Grade C");
}

</script>

</body>
</html>
```

Output

Grade A

Interview Questions

Q1. What is Control Flow?

Control Flow determines the order in which statements are executed in a program.

Q2. What is an if Statement?

The if statement executes a block of code only when a condition evaluates to true.

Q3. What are the types of if statements?

1. Simple if
2. if...else
3. else if Ladder
4. Nested if

Q4. When should we use switch?

Use switch when comparing one variable against multiple fixed values.

---

1. Simple if Statement Questions

Basic Level

Q1.

Check whether a number is positive.

Input: 10
Output: Positive Number

Q2.

Check whether a person is eligible for voting.

Input: Age = 20
Output: Eligible for Voting

Q3.

Check whether a student has passed.

Input: Marks = 40
Output: Pass

Q4.

Check whether a number is greater than 100.

Intermediate Level

Q5.

Check whether a character is uppercase.

Input: A
Output: Uppercase Letter

Q6.

Check whether a number is divisible by 5.

Q7.

Check whether a year is greater than 2000.

Q8.

Check whether a salary is above ₹50,000.

2. if...else Questions

Basic Level

Q1.

Check whether a number is even or odd.

Q2.

Check whether a number is positive or negative.

Q3.

Check whether a student passed or failed.

Pass Marks = 33

Q4.

Check whether a person is adult or minor.

Intermediate Level

Q5.

Find the larger number between two numbers.

Input: 10, 20
Output: 20 is Greater

Q6.

Check whether a year is leap year or not.

Q7.

Check whether a character is vowel or consonant.

Q8.

Check whether a number is divisible by both 3 and 5.

High Level

Q9.

Check whether a person is eligible for a driving license.

Conditions:

Age >=18
Indian Citizen

Q10.

Check whether a login password is correct.

---

3. else if Ladder Questions

Basic Level

Q1.

Display grades based on marks.

90+   A+
80+   A
70+   B
60+   C
Below 60 Fail

Q2.

Display day name.

1 = Monday
2 = Tuesday
3 = Wednesday

Q3.

Display month name using month number.

Intermediate Level

Q4.

Create a simple calculator.

+
-
*
/

Q5.

Find the largest among three numbers.

a=10
b=20
c=30

Q6.

Electricity Bill Calculation

0-100 units ₹5/unit
101-200 units ₹8/unit
201-300 units ₹10/unit

High Level

Q7.

Income Tax Calculator

0-2.5L No Tax
2.5L-5L 5%
5L-10L 20%
Above 10L 30%

Q8.

Employee Bonus System

Experience >10 Years = 20%
Experience >5 Years = 10%
Else = 5%

Q9.

Student Scholarship Eligibility

Marks >90
Income <200000

---

4. Nested if Questions

Basic Level

Q1.

Check Login System

Username Correct?
   Password Correct?

Q2.

ATM PIN Verification

Card Inserted?
   PIN Correct?

Intermediate Level

Q3.

Check Driving License Eligibility

Age >=18
   Medical Test Passed

Q4.

College Admission System

Marks >=60
    Documents Verified

High Level

Q5.

Online Shopping Discount

Customer Premium?
    Purchase >5000

Q6.

Loan Approval System

Salary >30000
    Credit Score >700

Pro Level

Q7.

Company Recruitment System

Degree Available?
    Experience >2 Years?
        Technical Round Cleared?

Output:

Selected
Not Selected

---

5. switch Case Questions

Basic Level

Q1.

Print Day Name.

1 = Monday
2 = Tuesday

Q2.

Print Month Name.

Q3.

Print Subject Name.

1 = HTML
2 = CSS
3 = JavaScript

Intermediate Level

Q4.

Create Calculator using switch.

+
-

*
/
%

Q5.

Create Menu Driven Program.

1. Add Student
2. Update Student
3. Delete Student
4. Search Student

Q6.

Food Ordering System

1. Pizza
2. Burger
3. Sandwich

High Level

Q7.

ATM Menu

1. Withdraw
2. Deposit
3. Balance Inquiry
4. Mini Statement

Q8.

Railway Reservation Menu

1. Book Ticket
2. Cancel Ticket
3. Check Status

Pro Level

Q9.

Employee Management System

1. Add Employee
2. Update Employee
3. Delete Employee
4. Search Employee
5. Exit

Ultimate Practice Set (Interview + Logic Building)

1. Check whether a number is positive, negative, or zero.

2. Find the largest among 3 numbers.

3. Find the largest among 4 numbers.

4. Check leap year.

5. Check vowel/consonant.

6. Check even/odd without % operator.

7. Create a Grade System.

8. Create an ATM System.

9. Create a Login System.

10. Create a Bank Loan Approval System.

11. Create an Income Tax Calculator.

12. Create a Student Scholarship Portal.

13. Create an Online Shopping Discount System.

14. Create a Railway Reservation Menu.

15. Create a Restaurant Billing System.

JavaScript Loops

A **loop** is used to execute a block of code repeatedly until a condition becomes false.

Without loops:

```
document.write("Welcome<br>");
document.write("Welcome<br>");
document.write("Welcome<br>");
document.write("Welcome<br>");
document.write("Welcome<br>");
```

With a loop:

```
for(let i=1; i<=5; i++)
{
   document.write("Welcome<br>");
}
```

Both produce the same output, but the loop is much shorter.

Types of Loops in JavaScript

1. for Loop
2. while Loop
3. do...while Loop
4. Nested Loops
5. break
6. continue

---

1. for Loop

Used when the number of iterations is known.

Syntax

```
for(initialization; condition; increment/decrement)
{
   // code
}
```

Example 1: Print Numbers 1 to 10

```
<script>

for(let i=1; i<=10; i++)
{
   document.write(i + "<br>");
}

</script>
```

Output

1
2
3
4
5
6
7
8
9
10

Example 2: Print Even Numbers

```
<script>

for(let i=2; i<=20; i=i+2)
{
   document.write(i + "<br>");
}

</script>
```

Output

2
4
6
8
10
12
14
16
18
20

Example 3: Reverse Counting

```
<script>

for(let i=10; i>=1; i--)
{
```

```
    document.write(i + "<br>");
}

</script>
```

Output

10
9
8
7
6
5
4
3
2
1

Ques 1. Create 10 Buttons Dynamically

```
<!DOCTYPE html>
<html>
<body>

<div id="buttons"></div>

<script>

let output = "";

for(let i=1; i<=10; i++)
{
```

```
   output += `<button>Button ${i}</button> `;
}

document.getElementById("buttons").innerHTML = output;

</script>

</body>
</html>
```

2. Generate Product Cards

```
<!DOCTYPE html>
<html>
<body>

<div id="products"></div>

<script>

let products =
[
   "Laptop",
   "Mobile",
   "Tablet",
   "Smart Watch"
];

let html = "";
for(let i=0; i<products.length; i++)
{
```

```
    html += `
    <div style="border:1px solid black;padding:10px;margin:10px;">
        <h3>${products[i]}</h3>
        <button>Buy Now</button>
    </div>`;
}
document.getElementById("products").innerHTML = html;
</script>
</body>
</html>
```

3. Dynamic Table Generation

```
<!DOCTYPE html>

<html>

<body>

<table border="1" id="table"></table>

<script>

let html = "";

for(let i=1; i<=10; i++)

{

    html += `

    <tr>

        <td>${i}</td>

        <td>Student ${i}</td>
```

```
    </tr>`;
}
document.getElementById("table").innerHTML = html;
</script>
</body>
</html>
```

4. Create Navigation Menu

```
<!DOCTYPE html>
<html>
<body>
<ul id="menu"></ul>
<script>
let items =
[
    "Home",
    "About",
    "Services",
    "Gallery",
```

```
    "Contact"
];
let html = "";
for(let i=0; i<items.length; i++)
{
    html += `<li>${items[i]}</li>`;
}
document.getElementById("menu").innerHTML = html;
</script>
</body>
</html>
```

5. Multiplication Table Generator

```
<!DOCTYPE html>
<html>
<body>
<script>
let num = 7;
for(let i=1; i<=10; i++)
```

```
{
    document.write(
    `${num} × ${i} = ${num*i}<br>`
    );
}
</script>
</body>
</html>
```

6. Star Rating System

```
<!DOCTYPE html>
<html>
<body>
<div id="rating"></div>
<script>
let stars = "";
for(let i=1; i<=5; i++)
{
    stars += "□";
```

```
}
document.getElementById("rating").innerHTML = stars;
</script>
</body>
</html>
```

7. Image Gallery

```
<!DOCTYPE html>
<html>
<body>
<div id="gallery"></div>
<script>
let html = "";
for(let i=1; i<=6; i++)
{
   html += `
   <img src="https://picsum.photos/200?random=${i}">
   `;
}
```

```
document.getElementById("gallery").innerHTML = html;
</script>
</body>
</html>
```

8. Countdown Timer Display

```
<!DOCTYPE html>
<html>
<body>
<script>
for(let i=10; i>=1; i--)
{
   document.write(i + "<br>");
}

document.write("⍰ Launch");
</script>
</body>
</html>
```

9. Generate Chess Board

```
<!DOCTYPE html>

<html>

<body>


<script>


for(let row=1; row<=8; row++)

{

   for(let col=1; col<=8; col++)

   {

      document.write("□ ");

   }


   document.write("<br>");

}

</script>

</body>
```

```
</html>
```

10. Student Result Dashboard

```
<!DOCTYPE html>
<html>
<body>
<script>
let marks = [78,85,92,65,88];
for(let i=0; i<marks.length; i++)
{
    document.write(
    `Student ${i+1} : ${marks[i]}<br>`
    );
}
</script>
</body>
</html>
```

11. Progress Bar Generator

```
<!DOCTYPE html>

<html>

<body>

<script>

let progress = "";

for(let i=1; i<=10; i++)

{

    progress += "█";

}

document.write(progress);

</script>

</body>

</html>
```

12. Dynamic Select Dropdown

```
<!DOCTYPE html>

<html>

<body>

<select id="cities"></select>
```

```
<script>
let city =
[
   "Jaipur",
   "Delhi",
   "Mumbai",
   "Pune"
];
let html = "";
for(let i=0; i<city.length; i++)
{
   html += `
   <option>${city[i]}</option>
   `;
}
document.getElementById("cities").innerHTML = html;
</script>
</body>
</html>
```

2. while Loop

Used when the number of iterations is not known in advance.

Syntax

```
while(condition)
{
   // code
}
```

Example

```
<script>

let i = 1;

while(i <= 5)
{
   document.write(i + "<br>");
   i++;
}

</script>
```

Output

1
2
3
4
5

Top While Loop Questions with Best Solutions (JavaScript)

These are the most important **lab, viva, interview, and logic-building questions** on while loops.

---

1. Print Numbers 1 to 10

```
let i = 1;

while(i <= 10)
{
   console.log(i);
   i++;
}
```

Output:

1 2 3 4 5 6 7 8 9 10

---

2. Print Even Numbers

```
let i = 2;

while(i <= 20)
{
   console.log(i);
   i += 2;
}
```

Output:

2 4 6 8 10 12 14 16 18 20

---

3. Print Odd Numbers

```
let i = 1;

while(i <= 20)
{
   console.log(i);
   i += 2;
}
```

---

4. Reverse Counting

```
let i = 10;

while(i >= 1)
{
   console.log(i);
   i--;
}
```

Output:

10 9 8 7 6 5 4 3 2 1

---

5. Sum of First 100 Numbers

```
let i = 1;
let sum = 0;

while(i <= 100)
```

```
{
   sum += i;
   i++;
}

console.log(sum);
```

Output:

5050

---

6. Multiplication Table

```
let num = 7;
let i = 1;

while(i <= 10)
{
   console.log(`${num} x ${i} = ${num*i}`);
   i++;
}
```

---

7. Count Digits in a Number ?

```
let num = 12345;
let count = 0;

while(num > 0)
{
   count++;
   num = Math.floor(num / 10);
```

```
}

console.log("Digits =", count);
```

Output:

Digits = 5

---

8. Sum of Digits ?

```
let num = 1234;
let sum = 0;

while(num > 0)
{
   sum += num % 10;
   num = Math.floor(num / 10);
}

console.log(sum);
```

Output:

10

---

9. Reverse a Number ?

```
let num = 1234;
let reverse = 0;

while(num > 0)
{
```

```
    let digit = num % 10;

    reverse = reverse * 10 + digit;

    num = Math.floor(num / 10);
}

console.log(reverse);
```

Output:

4321

---

10. Palindrome Number ? ? ?

```
let num = 121;
let temp = num;
let reverse = 0;

while(temp > 0)
{
    let digit = temp % 10;

    reverse = reverse * 10 + digit;

    temp = Math.floor(temp / 10);
}

if(reverse === num)
{
    console.log("Palindrome");
}
```

```
else
{
   console.log("Not Palindrome");
}
```

Output:

Palindrome

---

11. Armstrong Number

```
let num = 153;
let temp = num;
let sum = 0;

while(temp > 0)
{
   let digit = temp % 10;

   sum += digit ** 3;

   temp = Math.floor(temp / 10);
}

if(sum === num)
{
   console.log("Armstrong");
}
else
{
   console.log("Not Armstrong");
}
```

Output:

Armstrong

---

12. Factorial ⍰ ⍰ ⍰

```
let n = 5;
let fact = 1;

while(n > 0)
{
   fact *= n;
   n--;
}

console.log(fact);
```

Output:

120

---

13. Fibonacci Series ⍰ ⍰ ⍰

```
let a = 0;
let b = 1;
let i = 1;

while(i <= 10)
{
   console.log(a);

   let c = a + b;
```

```
    a = b;
    b = c;

    i++;
}
```

Output:

0 1 1 2 3 5 8 13 21 34

---

14. Prime Number ? ? ? ?

```
let num = 13;
let i = 2;
let prime = true;

while(i < num)
{
    if(num % i === 0)
    {
        prime = false;
        break;
    }

    i++;
}

console.log(
prime
? "Prime"
```

```
: "Not Prime"
);
```

Output:

Prime

---

15. Guess Number Game

```
let secret = 7;
let guess = 0;

while(guess != secret)
{
   guess = Number(
      prompt("Guess Number")
   );
}

alert("Correct!");
```

Real-world use:

- Games
- Login attempts
- Verification systems

---

16. ATM System

```
let balance = 10000;
let choice = "";
```

```
while(choice !== "4")
{
   choice = prompt(
`1.Deposit
2.Withdraw
3.Balance
4.Exit`
);

   if(choice === "1")
   {
      let amount =
      Number(prompt("Amount"));

      balance += amount;
   }

   else if(choice === "2")
   {
      let amount =
      Number(prompt("Amount"));

      balance -= amount;
   }

   else if(choice === "3")
   {
      alert(
      "Balance = " + balance
      );
   }
```

```
}

alert("Thank You");
```

---

17. Login Attempts System

```
let password = "1234";
let input = "";
let attempts = 3;

while(attempts > 0)
{
   input = prompt("Password");

   if(input === password)
   {
      alert("Login Success");
      break;
   }

   attempts--;

   alert(
   "Attempts Left: " +
   attempts
   );
}
```

---

18. Find Greatest Digit

```
let num = 59284;
let max = 0;

while(num > 0)
{
   let digit = num % 10;

   if(digit > max)
   {
      max = digit;
   }

   num = Math.floor(num / 10);
}

console.log(max);
```

Output:

9

---

19. Perfect Number

```
let num = 28;
let i = 1;
let sum = 0;

while(i < num)
{
   if(num % i === 0)
   {
      sum += i;
```

```
    }

    i++;
}

console.log(
sum === num
? "Perfect Number"
: "Not Perfect"
);
```

Output:

```
Perfect Number
```

---

20. Strong Number

```
let num = 145;
let temp = num;
let sum = 0;

while(temp > 0)
{
    let digit = temp % 10;

    let fact = 1;

    let i = 1;

    while(i <= digit)
    {
        fact *= i;
```

```
        i++;
    }

    sum += fact;

    temp = Math.floor(temp / 10);
}

console.log(
sum === num
? "Strong Number"
: "Not Strong"
);
```

Output:

```
Strong Number
```

Most Important While Loop Questions for Exams

Basic

1. Print 1–100
2. Even/Odd numbers
3. Reverse counting
4. Multiplication table
5. Sum of N numbers

Intermediate

6. Count digits
7. Sum of digits
8. Reverse number

9. Factorial
10. Fibonacci

High Level

11. Prime number
12. Armstrong number
13. Palindrome
14. Perfect number
15. Strong number

Real-World Projects

16. ATM System
17. Login System
18. Menu Driven Program
19. Guess Number Game
20. Shopping Cart System

Top While Loop Questions with Best Solutions (JavaScript)

These are the most important **lab, viva, interview, and logic-building questions** on while loops.

---

1. Print Numbers 1 to 10

```
let i = 1;

while(i <= 10)
{
   console.log(i);
   i++;
}
```

Output:

1 2 3 4 5 6 7 8 9 10

---

2. Print Even Numbers

```
let i = 2;

while(i <= 20)
{
   console.log(i);
   i += 2;
}
```

Output:

2 4 6 8 10 12 14 16 18 20

---

3. Print Odd Numbers

```
let i = 1;

while(i <= 20)
{
   console.log(i);
   i += 2;
}
```

---

4. Reverse Counting

```
let i = 10;
```

```
while(i >= 1)
{
   console.log(i);
   i--;
}
```

Output:

10 9 8 7 6 5 4 3 2 1

---

5. Sum of First 100 Numbers

```
let i = 1;
let sum = 0;

while(i <= 100)
{
   sum += i;
   i++;
}

console.log(sum);
```

Output:

5050

---

6. Multiplication Table

```
let num = 7;
let i = 1;
```

```
while(i <= 10)
{
   console.log(`${num} x ${i} = ${num*i}`);
   i++;
}
```

---

7. Count Digits in a Number

```
let num = 12345;
let count = 0;

while(num > 0)
{
   count++;
   num = Math.floor(num / 10);
}

console.log("Digits =", count);
```

Output:

Digits = 5

---

8. Sum of Digits

```
let num = 1234;
let sum = 0;

while(num > 0)
{
   sum += num % 10;
```

```
    num = Math.floor(num / 10);
}

console.log(sum);
```

Output:

10

---

9. Reverse a Number

```
let num = 1234;
let reverse = 0;

while(num > 0)
{
    let digit = num % 10;

    reverse = reverse * 10 + digit;

    num = Math.floor(num / 10);
}

console.log(reverse);
```

Output:

4321

---

10. Palindrome Number

```
let num = 121;
let temp = num;
let reverse = 0;

while(temp > 0)
{
   let digit = temp % 10;

   reverse = reverse * 10 + digit;

   temp = Math.floor(temp / 10);
}

if(reverse === num)
{
   console.log("Palindrome");
}
else
{
   console.log("Not Palindrome");
}
```

Output:

Palindrome

---

11. Armstrong Number

```
let num = 153;
let temp = num;
let sum = 0;
```

```
while(temp > 0)
{
   let digit = temp % 10;

   sum += digit ** 3;

   temp = Math.floor(temp / 10);
}

if(sum === num)
{
   console.log("Armstrong");
}
else
{
   console.log("Not Armstrong");
}
```

Output:

Armstrong

---

12. Factorial ? ? ?

```
let n = 5;
let fact = 1;

while(n > 0)
{
   fact *= n;
   n--;
}
```

```
console.log(fact);
```

Output:

120

---

13. Fibonacci Series

```
let a = 0;
let b = 1;
let i = 1;

while(i <= 10)
{
   console.log(a);

   let c = a + b;

   a = b;
   b = c;

   i++;
}
```

Output:

0 1 1 2 3 5 8 13 21 34

---

14. Prime Number

```
let num = 13;
let i = 2;
let prime = true;

while(i < num)
{
   if(num % i === 0)
   {
      prime = false;
      break;
   }

   i++;
}

console.log(
prime
? "Prime"
: "Not Prime"
);
```

Output:

```
Prime
```

---

15. Guess Number Game ? ? ? ?

```
let secret = 7;
let guess = 0;

while(guess != secret)
{
```

```
    guess = Number(
        prompt("Guess Number")
    );
}

alert("Correct!");
```

Real-world use:

- Games
- Login attempts
- Verification systems

---

16. ATM System

```
let balance = 10000;
let choice = "";

while(choice !== "4")
{
    choice = prompt(
`1.Deposit
2.Withdraw
3.Balance
4.Exit`
);

    if(choice === "1")
    {
        let amount =
        Number(prompt("Amount"));
```

```
        balance += amount;
    }

    else if(choice === "2")
    {
        let amount =
        Number(prompt("Amount"));

        balance -= amount;
    }

    else if(choice === "3")
    {
        alert(
        "Balance = " + balance
        );
    }
}

alert("Thank You");
```

---

17. Login Attempts System

```
let password = "1234";
let input = "";
let attempts = 3;

while(attempts > 0)
{
    input = prompt("Password");
```

```
    if(input === password)
    {
        alert("Login Success");
        break;
    }

    attempts--;

    alert(
    "Attempts Left: " +
    attempts
    );
}
```

---

18. Find Greatest Digit

```
let num = 59284;
let max = 0;

while(num > 0)
{
    let digit = num % 10;

    if(digit > max)
    {
        max = digit;
    }

    num = Math.floor(num / 10);
}
```

```
console.log(max);
```

Output:

9

---

19. Perfect Number

```
let num = 28;
let i = 1;
let sum = 0;

while(i < num)
{
   if(num % i === 0)
   {
      sum += i;
   }

   i++;
}

console.log(
sum === num
? "Perfect Number"
: "Not Perfect"
);
```

Output:

Perfect Number

20. Strong Number □ □ □ □ □

```
let num = 145;
let temp = num;
let sum = 0;

while(temp > 0)
{
   let digit = temp % 10;

   let fact = 1;

   let i = 1;

   while(i <= digit)
   {
      fact *= i;
      i++;
   }

   sum += fact;

   temp = Math.floor(temp / 10);
}

console.log(
sum === num
? "Strong Number"
: "Not Strong"
);
```

Output:

Strong Number

Most Important While Loop Questions for Exams

Basic

1. Print 1–100
2. Even/Odd numbers
3. Reverse counting
4. Multiplication table
5. Sum of N numbers

Intermediate

6. Count digits
7. Sum of digits
8. Reverse number
9. Factorial
10. Fibonacci

High Level

11. Prime number
12. Armstrong number
13. Palindrome
14. Perfect number
15. Strong number

Real-World Projects

16. ATM System
17. Login System

18. Menu Driven Program
19. Guess Number Game
20. Shopping Cart System

If you master these 20 programs, you'll cover almost every important concept of the while loop used in interviews and programming labs.

MORE INTERESTING QUESTION WITH ANSWER ON WHILE LOOP

20 More Interesting While Loop Questions with Answers (JavaScript)

These questions are excellent for **logic building, interviews, coding tests, and practical programming**.

---

1. Sum of Even Digits

Input:

123456

Output:

12

(2+4+6)

```
let num = 123456;
let sum = 0;

while(num > 0)
{
   let digit = num % 10;
```

```
    if(digit % 2 === 0)
    {
      sum += digit;
    }

    num = Math.floor(num / 10);
}

console.log(sum);
```

---

2. Product of Digits

```
let num = 1234;
let product = 1;

while(num > 0)
{
    product *= num % 10;
    num = Math.floor(num / 10);
}

console.log(product);
```

Output:

24

---

3. Count Even Digits

```
let num = 24681;
let count = 0;

while(num > 0)
{
   let digit = num % 10;

   if(digit % 2 === 0)
   {
      count++;
   }

   num = Math.floor(num / 10);
}

console.log(count);
```

Output:

3

---

4. Largest Digit in Number

```
let num = 92754;
let max = 0;

while(num > 0)
{
   let digit = num % 10;

   if(digit > max)
   {
```

```
        max = digit;
    }

    num = Math.floor(num / 10);
}

console.log(max);
```

Output:

9

---

5. Smallest Digit

```
let num = 92754;
let min = 9;

while(num > 0)
{
    let digit = num % 10;

    if(digit < min)
    {
        min = digit;
    }

    num = Math.floor(num / 10);
}

console.log(min);
```

Output:

2

---

6. Count Zeros

```
let num = 1002050;
let count = 0;

while(num > 0)
{
   if(num % 10 === 0)
   {
      count++;
   }

   num = Math.floor(num / 10);
}

console.log(count);
```

Output:

4

---

7. Binary to Decimal

```
let binary = 1011;
let power = 0;
let decimal = 0;

while(binary > 0)
{
```

```
    let digit = binary % 10;

    decimal += digit * (2 ** power);

    power++;

    binary = Math.floor(binary / 10);
}

console.log(decimal);
```

Output:

11

---

8. Decimal to Binary

```
let num = 13;
let binary = "";

while(num > 0)
{
    binary = (num % 2) + binary;
    num = Math.floor(num / 2);
}

console.log(binary);
```

Output:

1101

---

9. Power Calculation

Find:

2^5 = 32

```
let base = 2;
let exponent = 5;
let result = 1;

while(exponent > 0)
{
   result *= base;
   exponent--;
}

console.log(result);
```

---

10. Number Guessing Game

```
let secret = 25;
let guess = 0;

while(guess !== secret)
{
   guess = Number(prompt("Guess Number"));
}

alert("Correct Answer");
```

---

11. Password Verification

```
let password = "admin123";
let input = "";

while(input !== password)
{
   input = prompt("Enter Password");
}

alert("Access Granted");
```

---

12. Reverse String

```
let str = "javascript";
let i = str.length - 1;

while(i >= 0)
{
   document.write(str[i]);
   i--;
}
```

Output:

tpircsavaj

---

13. Count Vowels

```
let str = "javascript";
let i = 0;
let count = 0;
```

```
while(i < str.length)
{
   if("aeiou".includes(str[i]))
   {
      count++;
   }

   i++;
}

console.log(count);
```

Output:

3

---

14. Print Alphabet A-Z

```
let i = 65;

while(i <= 90)
{
   console.log(String.fromCharCode(i));
   i++;
}
```

Output:

A B C D ... Z

---

15. Generate Random OTP

```
let otp = "";

while(otp.length < 6)
{
   otp += Math.floor(Math.random()*10);
}

console.log(otp);
```

Output:

583921

---

16. ATM Cash Withdrawal

```
let balance = 10000;

while(balance > 0)
{
   let amount = Number(prompt("Withdraw Amount"));

   if(amount > balance)
   {
      alert("Insufficient Balance");
   }
   else
   {
      balance -= amount;
      alert("Remaining = " + balance);
   }

   if(balance === 0)
```

```
    {
        break;
    }
}
```

---

17. Multiplication Without *

```
let a = 5;
let b = 4;
let result = 0;

while(b > 0)
{
    result += a;
    b--;
}

console.log(result);
```

Output:

20

---

18. Division Without /

```
let dividend = 20;
let divisor = 5;
let count = 0;

while(dividend >= divisor)
{
```

```
    dividend -= divisor;
    count++;
}

console.log(count);
```

Output:

4

---

19. Find HCF (GCD)

```
let a = 24;
let b = 36;

while(b !== 0)
{
    let temp = b;

    b = a % b;

    a = temp;
}

console.log(a);
```

Output:

12

---

20. Find LCM

```
let a = 12;
let b = 15;

let max = (a > b) ? a : b;

while(true)
{
   if(max % a === 0 && max % b === 0)
   {
      console.log(max);
      break;
   }

   max++;
}
```

Output:

60

Pro-Level Challenge Questions (Try Yourself)

1. Check Happy Number
2. Check Neon Number
3. Check Duck Number
4. Check Spy Number
5. Check Automorphic Number
6. Find Nth Fibonacci Number
7. Generate Prime Numbers Between 1–1000
8. Generate Armstrong Numbers Between 1–10000
9. Student Login System with 3 Attempts
10. Restaurant Billing System

11. Online Shopping Cart
12. Railway Reservation Menu
13. Library Management Menu
14. Employee Payroll System
15. Quiz Application
16. E-commerce Checkout System
17. OTP Verification System
18. Voting System
19. Banking Menu System
20. Calculator Using While Loop

---

3. do...while Loop

Executes the block at least once, even if the condition is false.

Syntax

```
do
{
   // code
}
while(condition);
```

Example

```
<script>

let i = 1;

do
{
```

```
    document.write(i + "<br>");
    i++;
}
while(i <= 5);
</script>
```

Interesting do...while Loop Questions with Answers (JavaScript)

What is do...while?

A do...while loop executes the code **at least once**, even if the condition is false.

```
do {
   // code
} while(condition);
```

---

1. Print Numbers 1 to 10

```
let i = 1;

do {
   console.log(i);
   i++;
} while(i <= 10);
```

Output:

1 2 3 4 5 6 7 8 9 10

---

2. Print Even Numbers

```
let i = 2;

do {
  console.log(i);
  i += 2;
} while(i <= 20);
```

Output:

2 4 6 8 10 12 14 16 18 20

---

3. Multiplication Table

```
let num = 5;
let i = 1;

do {
  console.log(`${num} x ${i} = ${num*i}`);
  i++;
} while(i <= 10);
```

---

4. Reverse Counting

```
let i = 10;

do {
  console.log(i);
  i--;
} while(i >= 1);
```

Output:

10 9 8 7 6 5 4 3 2 1

---

5. Sum of Numbers 1 to 100

```
let i = 1;
let sum = 0;

do {
   sum += i;
   i++;
} while(i <= 100);

console.log(sum);
```

Output:

5050

---

6. Password Verification

```
let password = "admin123";
let input;

do {
   input = prompt("Enter Password");
} while(input !== password);

alert("Login Successful");
```

---

7. ATM PIN Verification

```
let pin = "1234";
let enteredPin;

do {
  enteredPin = prompt("Enter ATM PIN");
} while(enteredPin !== pin);

alert("Access Granted");
```

---

8. Number Guessing Game

```
let secret = 25;
let guess;

do {
  guess = Number(prompt("Guess Number"));
} while(guess !== secret);

alert("Correct Answer");
```

---

9. Generate OTP

```
let otp = "";

do {
  otp += Math.floor(Math.random() * 10);
} while(otp.length < 6);

console.log(otp);
```

Output Example:

483921

---

10. Menu Driven Calculator

```
let choice;

do {
   choice = prompt(`
1. Add
2. Exit
`);

   if(choice == 1)
   {
      let a = Number(prompt("Enter A"));
      let b = Number(prompt("Enter B"));

      alert(a+b);
   }

} while(choice != 2);
```

---

11. Reverse a Number

```
let num = 1234;
let reverse = 0;

do {
   let digit = num % 10;

   reverse = reverse * 10 + digit;
```

```
    num = Math.floor(num / 10);

} while(num > 0);

console.log(reverse);
```

Output:

4321

---

12. Count Digits

```
let num = 987654;
let count = 0;

do {
    count++;
    num = Math.floor(num / 10);

} while(num > 0);

console.log(count);
```

Output:

6

---

13. Sum of Digits

```
let num = 12345;
let sum = 0;
```

```
do {
   sum += num % 10;

   num = Math.floor(num / 10);

} while(num > 0);

console.log(sum);
```

Output:

15

---

14. Armstrong Number

```
let num = 153;
let temp = num;
let sum = 0;

do {
   let digit = temp % 10;

   sum += digit ** 3;

   temp = Math.floor(temp / 10);

} while(temp > 0);

console.log(
sum === num
? "Armstrong"
```

```
: "Not Armstrong"
);
```

Output:

Armstrong

---

15. Palindrome Number

```
let num = 121;
let temp = num;
let reverse = 0;

do {
   let digit = temp % 10;

   reverse = reverse * 10 + digit;

   temp = Math.floor(temp / 10);

} while(temp > 0);

console.log(
reverse === num
? "Palindrome"
: "Not Palindrome"
);
```

---

16. Fibonacci Series

```
let a = 0;
let b = 1;
let count = 1;

do {
   console.log(a);

   let c = a + b;

   a = b;
   b = c;

   count++;

} while(count <= 10);
```

Output:

0 1 1 2 3 5 8 13 21 34

---

17. Shopping Cart System

```
let total = 0;
let choice;

do {
   let price =
   Number(prompt("Enter Product Price"));

   total += price;

   choice =
```

```
    prompt("Add More? yes/no");

} while(choice === "yes");

alert("Total Bill = " + total);
```

---

18. Restaurant Billing System

```
let bill = 0;
let item;

do {
    item = Number(
    prompt("Enter Item Price")
    );

    bill += item;

} while(
confirm("Add Another Item?")
);

alert("Total Bill = " + bill);
```

---

19. Student Marks Average

```
let total = 0;
let count = 0;
let marks;

do {
```

```
    marks =
    Number(prompt("Enter Marks"));

    total += marks;
    count++;

} while(confirm("Add More Marks?"));

console.log(
"Average = " + (total/count)
);
```

---

20. Mini ATM System (Most Interesting)

```
let balance = 10000;
let choice;

do {

    choice = prompt(`
1. Balance
2. Deposit
3. Withdraw
4. Exit
`);

    switch(choice)
    {
        case "1":
            alert(
            "Balance = " + balance
```

```
        );
        break;

      case "2":
        balance += Number(
        prompt("Amount")
        );
        break;

      case "3":
        balance -= Number(
        prompt("Amount")
        );
        break;
    }

} while(choice !== "4");

alert("Thank You");
```

Most Asked Interview Questions on do...while

Basic

1. Print 1–100
2. Even/Odd Numbers
3. Multiplication Table
4. Sum of N Numbers
5. Reverse Counting

Intermediate

6. Count Digits
7. Reverse Number
8. Sum of Digits
9. Palindrome
10. Armstrong Number

Advanced

11. Password Verification
12. Login System
13. OTP Verification
14. ATM System
15. Shopping Cart

Real Projects

16. Billing System
17. Quiz Application
18. Voting System
19. Railway Reservation Menu
20. Banking Menu System

Special Interview Question

Why use do...while instead of while?

```
let num = 10;

do {
  console.log("Executed");
} while(num < 5);
```

Difference Between while and do...while

```
let i = 10;

while(i <= 5)
{
   document.write("Hello");
}
```

Output:

Nothing

```
let i = 10;

do
{
   document.write("Hello");
}
while(i <= 5);
```

Output:

Hello

---

break Statement

Used to stop a loop immediately.

Example

```
<script>

for(let i=1; i<=10; i++)
```

```
{
   if(i==5)
   {
      break;
   }

   document.write(i + "<br>");
}

</script>
```

Output

1
2
3
4

---

continue Statement

Used to skip the current iteration.

Example

```
<script>

for(let i=1; i<=10; i++)
{
   if(i==5)
   {
      continue;
   }
```

```
    document.write(i + "<br>");
}

</script>
```

Output

1
2
3
4
6
7
8
9
10

Practical Programs

Program 1: Sum of 1 to 10

```
<script>

let sum = 0;

for(let i=1; i<=10; i++)
{
    sum += i;
}

document.write("Sum = " + sum);
```

```
</script>
```

Output

Sum = 55

Program 2: Multiplication Table

```
<script>

let num = 5;

for(let i=1; i<=10; i++)
{
   document.write(num + " x " + i + " = " + (num*i) + "<br>");
}

</script>
```

Output

5 x 1 = 5
5 x 2 = 10
...
5 x 10 = 50

Program 3: Factorial

Formula

n!=n×(n−1)×(n−2)⋯1n! = n \times (n-1) \times (n-2) \cdots 1n!=n×(n−1)×(n−2)⋯1

```
<script>

let n = 5;
let fact = 1;

for(let i=1; i<=n; i++)
{
   fact *= i;
}

document.write("Factorial = " + fact);

</script>
```

Output

Factorial = 120

Program 4: Count Digits

```
<script>

let num = 12345;
let count = 0;

while(num > 0)
{
   count++;
   num = Math.floor(num / 10);
}

document.write("Digits = " + count);
```

```
</script>
```

Output

Digits = 5

Program 5: Sum of Digits

```
<script>

let num = 123;
let sum = 0;

while(num > 0)
{
   sum += num % 10;
   num = Math.floor(num / 10);
}

document.write("Sum = " + sum);

</script>
```

Output

Sum = 6

Nested Loop

A loop inside another loop.

Example

```
<script>

for(let i=1; i<=3; i++)
{
   for(let j=1; j<=3; j++)
   {
      document.write("* ");
   }

   document.write("<br>");
}

</script>
```

Output

```
* * *
* * *
* * *
```

Interview Questions

Q1. Difference between for and while?

for	while
Known iterations	Unknown iterations
Compact syntax	Flexible

Q2. Difference between while and do...while?

while	do...while

while	do...while
Checks condition first	Executes once before checking
May run 0 times	Runs at least 1 time

Q3. What does break do?

Stops the loop immediately.

Q4. What does continue do?

Skips the current iteration and moves to the next.

Practice Questions

Basic

1. Print 1 to 100.
2. Print 100 to 1.
3. Print even numbers from 1 to 50.
4. Print odd numbers from 1 to 50.
5. Print multiplication table of 7.

Intermediate

6. Find sum of first 100 numbers.
7. Find factorial of a number.
8. Count digits in a number.
9. Reverse a number.
10. Sum of digits.

High Level

11. Check palindrome number.
12. Check Armstrong number.
13. Check prime number.
14. Generate Fibonacci series.
15. Find greatest digit in a number.

Pro Level

16. Pattern programs (*, numbers, pyramids).
17. Prime numbers between 1 and 100.
18. Perfect number.
19. Strong number.
20. Menu-driven calculator using loops.

After mastering loops, the next topic should be **Functions in JavaScript**, which is one of the most important topics for interviews and real-world programming.

also explain advance loops

After learning **for, while, do...while**, the next level is **Advanced Loops in JavaScript**.

Advanced Loops in JavaScript

A. Nested Loops
B. for...in Loop
C. for...of Loop
D. forEach()
E. map()
F. filter()
G. reduce()
H. find()
I. some()

J. every()

---

1. Nested Loops

A loop inside another loop.

```
for(let i=1; i<=3; i++)
{
   for(let j=1; j<=3; j++)
   {
      document.write("* ");
   }
   document.write("<br>");
}
```

Output

```
* * *
* * *
* * *
```

Pattern Program

```
for(let i=1; i<=5; i++)
{
   for(let j=1; j<=i; j++)
   {
      document.write("* ");
   }
   document.write("<br>");
}
```

Output

```
*
* *
* * *
* * * *
* * * * *
```

---

2. for...in Loop

Used to iterate over **object properties**.

```
let student =
{
   name:"Vicky",
   age:25,
   city:"Jaipur"
};

for(let key in student)
{
   console.log(key);
}
```

Output

```
name
age
city
```

Print Keys and Values

```
let student =
{
   name:"Vicky",
   age:25,
   city:"Jaipur"
};

for(let key in student)
{
   console.log(key + " : " + student[key]);
}
```

Output

```
name : Vicky
age : 25
city : Jaipur
```

1. Print All Student Details

```
let student = {
   name: "Vicky",
   age: 25,
   course: "BCA",
   city: "Jaipur"
};

for(let key in student)
{
   console.log(key + " : " + student[key]);
}
```

Output

name : Vicky
age : 25
course : BCA
city : Jaipur

---

2. Count Number of Properties

```
let employee = {
   id:101,
   name:"Rahul",
   salary:50000,
   department:"IT"
};

let count = 0;

for(let key in employee)
{
   count++;
}

console.log("Total Properties =", count);
```

Output

Total Properties = 4

---

3. Sum All Numeric Values

```
let marks = {
   html:80,
```

```
   css:85,
   js:90,
   react:95
};

let total = 0;

for(let key in marks)
{
   total += marks[key];
}

console.log("Total =", total);
```

Output

Total = 350

---

4. Find Highest Marks

```
let marks = {
   html:80,
   css:85,
   js:95,
   react:90
};

let max = 0;

for(let subject in marks)
{
   if(marks[subject] > max)
```

```
    {
        max = marks[subject];
    }
}

console.log("Highest Marks =", max);
```

Output

Highest Marks = 95

---

5. Print Only Keys

```
let car = {
    brand:"BMW",
    model:"X5",
    price:8000000
};

for(let key in car)
{
    console.log(key);
}
```

Output

brand
model
price

---

6. Print Only Values

```
let car = {
   brand:"BMW",
   model:"X5",
   price:8000000
};

for(let key in car)
{
   console.log(car[key]);
}
```

Output

```
BMW
X5
8000000
```

---

7. Find Property Exists or Not

```
let student = {
   name:"Vicky",
   age:25,
   city:"Jaipur"
};

let found = false;

for(let key in student)
{
   if(key === "city")
   {
      found = true;
```

```
        break;
    }
}

console.log(found ? "Found" : "Not Found");
```

Output

Found

---

8. Create Dynamic HTML Table

```
<div id="result"></div>

<script>

let student = {
    name:"Vicky",
    age:25,
    city:"Jaipur"
};

let html = "<table border='1'>";

for(let key in student)
{
    html += `
    <tr>
        <td>${key}</td>
        <td>${student[key]}</td>
    </tr>`;
}
```

```
html += "</table>";

document.getElementById("result").innerHTML = html;

</script>
```

Output

```
+------+--------+
| name | Vicky  |
| age  | 25     |
| city | Jaipur |
+------+--------+
```

---

9. Count String Properties

```
let data = {
   name:"Vicky",
   age:25,
   city:"Jaipur",
   salary:50000
};

let count = 0;

for(let key in data)
{
   if(typeof data[key] === "string")
   {
      count++;
   }
```

```
}

console.log(count);
```

Output

2

---

10. Find Longest String Value

```
let user = {
   firstName:"Vicky",
   city:"Jaipur",
   profession:"Computer Professor"
};

let longest = "";

for(let key in user)
{
   if(typeof user[key] === "string" &&
      user[key].length > longest.length)
   {
       longest = user[key];
   }
}

console.log(longest);
```

Output

Computer Professor

11. Convert Object to Array

```
let student = {
   name:"Vicky",
   age:25,
   city:"Jaipur"
};

let arr = [];

for(let key in student)
{
   arr.push(student[key]);
}

console.log(arr);
```

Output

```
["Vicky",25,"Jaipur"]
```

---

12. Shopping Cart Total

```
let cart = {
   laptop:50000,
   mobile:20000,
   mouse:1000
};

let total = 0;
```

```
for(let item in cart)
{
   total += cart[item];
}

console.log("Bill =", total);
```

Output

Bill = 71000

---

13. Employee Salary Increment

```
let employee = {
   Rahul:30000,
   Aman:40000,
   Neha:50000
};

for(let name in employee)
{
   employee[name] += 5000;
}

console.log(employee);
```

Output

```
{
 Rahul:35000,
 Aman:45000,
```

```
 Neha:55000
}
```

---

14. Print Nested Object

```
let student = {
   name:"Vicky",
   marks:{
      html:80,
      css:90,
      js:95
   }
};

for(let key in student.marks)
{
   console.log(
   key + " = " +
   student.marks[key]
   );
}
```

Output

```
html = 80
css = 90
js = 95
```

---

15. Real-World Product Listing

```
let products = {
   Laptop:50000,
   Mobile:25000,
   Tablet:15000
};

for(let product in products)
{
   console.log(
   product +
   " ₹" +
   products[product]
   );
}
```

Output

```
Laptop ₹50000
Mobile ₹25000
Tablet ₹15000
```

Pro-Level Interview Questions

16. Find Highest Salary Employee

```
let employees = {
   Rahul:30000,
   Aman:45000,
   Neha:60000,
   Riya:55000
};
```

```
let highest = 0;
let empName = "";

for(let name in employees)
{
   if(employees[name] > highest)
   {
      highest = employees[name];
      empName = name;
   }
}

console.log(empName, highest);
```

Output

Neha 60000

---

17. Count Boolean Values

```
let settings = {
   darkMode:true,
   notifications:false,
   sound:true
};

let count = 0;

for(let key in settings)
{
   if(typeof settings[key] === "boolean")
   {
```

```
        count++;
    }
}

console.log(count);
```

Output

3

---

18. Remove Empty Values

```
let user = {
    name:"Vicky",
    email:"",
    city:"Jaipur"
};

for(let key in user)
{
    if(user[key] === "")
    {
        delete user[key];
    }
}

console.log(user);
```

Output

```
{
 name:"Vicky",
```

```
 city:"Jaipur"
}
```

Most Useful Real-World Uses of for...in

1. Form Validation

```
for(let field in formData)
{
   if(formData[field] === "")
   {
      console.log(field + " Required");
   }
}
```

2. API Response Processing

```
for(let key in response)
{
   console.log(key, response[key]);
}
```

3. User Profile Display

```
for(let key in user)
{
   document.write(
   `${key}: ${user[key]}<br>`
   );
}
```

4. Dynamic Tables

```
for(let key in data)
{
  // create rows dynamically
}
```

Practice Set

1. Find Lowest Marks
2. Find Average Marks
3. Count Numeric Properties
4. Count String Properties
5. Count Boolean Properties
6. Print Object in HTML Table
7. Find Longest String
8. Find Highest Salary Employee
9. Shopping Cart Bill Calculator
10. Student Report Card Generator

---

3. for...of Loop

Used for Arrays and Strings.

```
let fruits = ["Apple","Mango","Orange"];

for(let fruit of fruits)
{
   console.log(fruit);
}
```

Output

Apple
Mango
Orange

String Example

```
let name = "Vicky";

for(let ch of name)
{
   console.log(ch);
}
```

Output

```
V
i
c
k
y
```

1. Print All Student Details

```
let student = {
   name: "Vicky",
   age: 25,
   course: "BCA",
   city: "Jaipur"
};

for(let key in student)
```

```
{
   console.log(key + " : " + student[key]);
}
```

Output

```
name : Vicky
age : 25
course : BCA
city : Jaipur
```

---

2. Count Number of Properties

```
let employee = {
   id:101,
   name:"Rahul",
   salary:50000,
   department:"IT"
};

let count = 0;

for(let key in employee)
{
   count++;
}

console.log("Total Properties =", count);
```

Output

```
Total Properties = 4
```

---

3. Sum All Numeric Values

```
let marks = {
   html:80,
   css:85,
   js:90,
   react:95
};

let total = 0;

for(let key in marks)
{
   total += marks[key];
}

console.log("Total =", total);
```

Output

Total = 350

---

4. Find Highest Marks

```
let marks = {
   html:80,
   css:85,
   js:95,
   react:90
};
```

```
let max = 0;

for(let subject in marks)
{
   if(marks[subject] > max)
   {
      max = marks[subject];
   }
}

console.log("Highest Marks =", max);
```

Output

```
Highest Marks = 95
```

---

5. Print Only Keys

```
let car = {
   brand:"BMW",
   model:"X5",
   price:8000000
};

for(let key in car)
{
   console.log(key);
}
```

Output

brand
model
price

---

6. Print Only Values

```
let car = {
   brand:"BMW",
   model:"X5",
   price:8000000
};

for(let key in car)
{
   console.log(car[key]);
}
```

Output

BMW
X5
8000000

---

7. Find Property Exists or Not

```
let student = {
   name:"Vicky",
   age:25,
   city:"Jaipur"
};
```

```
let found = false;

for(let key in student)
{
    if(key === "city")
    {
        found = true;
        break;
    }
}

console.log(found ? "Found" : "Not Found");
```

Output

Found

---

8. Create Dynamic HTML Table

```
<div id="result"></div>

<script>

let student = {
    name:"Vicky",
    age:25,
    city:"Jaipur"
};

let html = "<table border='1'>";

for(let key in student)
```

```
{
   html += `
   <tr>
      <td>${key}</td>
      <td>${student[key]}</td>
   </tr>`;
}

html += "</table>";

document.getElementById("result").innerHTML = html;

</script>
```

Output

```
+------+--------+
| name | Vicky  |
| age  | 25     |
| city | Jaipur |
+------+--------+
```

---

9. Count String Properties

```
let data = {
   name:"Vicky",
   age:25,
   city:"Jaipur",
   salary:50000
};

let count = 0;
```

```
for(let key in data)
{
    if(typeof data[key] === "string")
    {
        count++;
    }
}

console.log(count);
```

Output

2

---

10. Find Longest String Value

```
let user = {
    firstName:"Vicky",
    city:"Jaipur",
    profession:"Computer Professor"
};

let longest = "";

for(let key in user)
{
    if(typeof user[key] === "string" &&
       user[key].length > longest.length)
    {
        longest = user[key];
    }
```

```
}

console.log(longest);
```

Output

Computer Professor

---

11. Convert Object to Array

```
let student = {
   name:"Vicky",
   age:25,
   city:"Jaipur"
};

let arr = [];

for(let key in student)
{
   arr.push(student[key]);
}

console.log(arr);
```

Output

["Vicky",25,"Jaipur"]

---

12. Shopping Cart Total

```
let cart = {
   laptop:50000,
   mobile:20000,
   mouse:1000
};

let total = 0;

for(let item in cart)
{
   total += cart[item];
}

console.log("Bill =", total);
```

Output

```
Bill = 71000
```

---

13. Employee Salary Increment

```
let employee = {
   Rahul:30000,
   Aman:40000,
   Neha:50000
};

for(let name in employee)
{
   employee[name] += 5000;
}
```

```
console.log(employee);
```

Output

```
{
 Rahul:35000,
 Aman:45000,
 Neha:55000
}
```

---

14. Print Nested Object

```
let student = {
   name:"Vicky",
   marks:{
      html:80,
      css:90,
      js:95
   }
};

for(let key in student.marks)
{
   console.log(
   key + " = " +
   student.marks[key]
   );
}
```

Output

```
html = 80
css = 90
js = 95
```

---

15. Real-World Product Listing

```
let products = {
   Laptop:50000,
   Mobile:25000,
   Tablet:15000
};

for(let product in products)
{
   console.log(
   product +
   " ₹" +
   products[product]
   );
}
```

Output

```
Laptop ₹50000
Mobile ₹25000
Tablet ₹15000
```

Pro-Level Interview Questions

16. Find Highest Salary Employee

```
let employees = {
   Rahul:30000,
   Aman:45000,
   Neha:60000,
   Riya:55000
};

let highest = 0;
let empName = "";

for(let name in employees)
{
   if(employees[name] > highest)
   {
      highest = employees[name];
      empName = name;
   }
}

console.log(empName, highest);
```

Output

Neha 60000

---

17. Count Boolean Values

```
let settings = {
   darkMode:true,
   notifications:false,
   sound:true
};
```

```
let count = 0;

for(let key in settings)
{
   if(typeof settings[key] === "boolean")
   {
      count++;
   }
}

console.log(count);
```

Output

```
3
```

---

18. Remove Empty Values

```
let user = {
   name:"Vicky",
   email:"",
   city:"Jaipur"
};

for(let key in user)
{
   if(user[key] === "")
   {
      delete user[key];
   }
}
```

```
console.log(user);
```

Output

```
{
 name:"Vicky",
 city:"Jaipur"
}
```

Most Useful Real-World Uses of for...in

1. Form Validation

```
for(let field in formData)
{
   if(formData[field] === "")
   {
      console.log(field + " Required");
   }
}
```

2. API Response Processing

```
for(let key in response)
{
   console.log(key, response[key]);
}
```

3. User Profile Display

```
for(let key in user)
{
   document.write(
```

```
    `${key}: ${user[key]}<br>`
  );
}
```

4. Dynamic Tables

```
for(let key in data)
{
  // create rows dynamically
}
```

Practice Set

1. Find Lowest Marks
2. Find Average Marks
3. Count Numeric Properties
4. Count String Properties
5. Count Boolean Properties
6. Print Object in HTML Table
7. Find Longest String
8. Find Highest Salary Employee
9. Shopping Cart Bill Calculator
10. Student Report Card Generator

Difference Between for...in and for...of

for...in	for...of
Object Keys	Values
Objects	Arrays, Strings

Example

```
let arr = ["HTML","CSS","JS"];

for(let i in arr)
{
   console.log(i);
}
```

Output

```
0
1
2
```

```
for(let value of arr)
{
   console.log(value);
}
```

Output

```
HTML
CSS
JS
```

---

4. forEach()

Used with arrays.

```
let nums = [10,20,30,40];

nums.forEach(function(value)
```

```
{
   console.log(value);
});
```

Output

```
10
20
30
40
```

Arrow Function Version

```
let nums = [10,20,30];

nums.forEach(num =>
{
   console.log(num);
});
```

JavaScript forEach() Loop - Important Questions with Answers

What is forEach()?

forEach() is an array method that executes a function for each array element.

Syntax

```
array.forEach(function(value, index, array)
{
   // code
});
```

Example

```
let fruits = ["Apple", "Mango", "Orange"];

fruits.forEach(function(item)
{
   console.log(item);
});
```

Output:

```
Apple
Mango
Orange
```

---

1. Print All Array Elements

```
let numbers = [10,20,30,40];

numbers.forEach(function(num)
{
   console.log(num);
});
```

Output:

```
10
20
30
40
```

---

2. Sum of Array Elements

```
let numbers = [10,20,30,40];

let sum = 0;

numbers.forEach(function(num)
{
   sum += num;
});

console.log(sum);
```

Output:

100

---

3. Print Squares of Numbers

```
let numbers = [1,2,3,4,5];

numbers.forEach(function(num)
{
   console.log(num * num);
});
```

Output:

```
1
4
9
16
25
```

---

4. Print Even Numbers

```
let numbers = [10,11,12,13,14];

numbers.forEach(function(num)
{
   if(num % 2 === 0)
   {
      console.log(num);
   }
});
```

Output:

```
10
12
14
```

---

5. Count Even Numbers

```
let numbers = [10,11,12,13,14];

let count = 0;

numbers.forEach(function(num)
{
   if(num % 2 === 0)
   {
      count++;
   }
});
```

```
console.log(count);
```

Output:

3

---

6. Find Maximum Number

```
let numbers = [25,50,10,80,40];

let max = numbers[0];

numbers.forEach(function(num)
{
   if(num > max)
   {
      max = num;
   }
});

console.log(max);
```

Output:

80

---

7. Find Minimum Number

```
let numbers = [25,50,10,80,40];

let min = numbers[0];
```

```
numbers.forEach(function(num)
{
   if(num < min)
   {
      min = num;
   }
});

console.log(min);
```

Output:

10

---

8. Add 10 to Every Element

```
let numbers = [10,20,30];

numbers.forEach(function(num)
{
   console.log(num + 10);
});
```

Output:

20
30
40

---

9. Print Index and Value

```
let fruits = ["Apple","Mango","Orange"];

fruits.forEach(function(value,index)
{
   console.log(index + " = " + value);
});
```

Output:

```
0 = Apple
1 = Mango
2 = Orange
```

---

10. Convert Names to Uppercase

```
let names = ["rahul","aman","vicky"];

names.forEach(function(name)
{
   console.log(name.toUpperCase());
});
```

Output:

```
RAHUL
AMAN
VICKY
```

---

11. Calculate Shopping Cart Total

```
let cart = [50000,20000,1000,500];
```

```
let total = 0;

cart.forEach(function(price)
{
   total += price;
});

console.log("Total Bill =", total);
```

Output:

Total Bill = 71500

---

12. Student Marks Average

```
let marks = [80,90,70,85,95];

let total = 0;

marks.forEach(function(mark)
{
   total += mark;
});

console.log(total / marks.length);
```

Output:

84

---

13. Count Vowels in Array of Words

```
let words = ["apple","banana","orange"];

let count = 0;

words.forEach(function(word)
{
   for(let ch of word)
   {
      if("aeiou".includes(ch))
      {
         count++;
      }
   }
});

console.log(count);
```

Output:

8

---

14. Generate HTML List

```
let products = [
   "Laptop",
   "Mobile",
   "Tablet"
];

let html = "";

products.forEach(function(product)
```

```
{
   html += `<li>${product}</li>`;
});

document.body.innerHTML =
`<ul>${html}</ul>`;
```

Output:

- Laptop
- Mobile
- Tablet

---

15. Employee Salary Increment

```
let salaries =
[30000,40000,50000];

salaries.forEach(function(salary)
{
   console.log(
   salary + 5000
   );
});
```

Output:

35000
45000
55000

---

16. Print Table of 5

```
let table =
[1,2,3,4,5,6,7,8,9,10];

table.forEach(function(num)
{
   console.log(
   `5 x ${num} = ${5*num}`
   );
});
```

---

17. Product Discount System

```
let prices =
[1000,2000,3000];

prices.forEach(function(price)
{
   let discount =
   price * 0.10;

   console.log(
   price - discount
   );
});
```

Output:

```
900
1800
2700
```

---

18. Display User Cards

```
let users =
[
   "Rahul",
   "Aman",
   "Vicky"
];

users.forEach(function(user)
{
   document.write(
   `<h2>${user}</h2>`
   );
});
```

---

19. Count Positive Numbers

```
let numbers =
[-10,20,-5,30,40];

let count = 0;

numbers.forEach(function(num)
{
   if(num > 0)
   {
      count++;
   }
});
```

console.log(count);

Output:

3

---

20. Create Dynamic Product Cards

```
let products =
[
{
name:"Laptop",
price:50000
},
{
name:"Mobile",
price:25000
}
];

products.forEach(function(product)
{
   document.write(`
   <div>
      <h2>${product.name}</h2>
      <p>₹${product.price}</p>
   </div>
   `);
});
```

Most Important Interview Questions

Basic

1. Print Array Elements
2. Sum of Array
3. Count Even Numbers
4. Find Maximum Number
5. Find Minimum Number

Intermediate

6. Average Marks
7. Shopping Cart Total
8. Discount Calculator
9. Count Positive Numbers
10. Generate HTML List

Advanced

11. Product Cards
12. Employee Salary System
13. Student Report Card
14. User Dashboard
15. E-commerce Product Listing

Difference: forEach vs for...of

forEach

```
let arr = [10,20,30];

arr.forEach(function(num)
```

```
{
   console.log(num);
});
```

for...of

```
for(let num of arr)
{
   console.log(num);
}
```

Key Difference

Feature	forEach()	for...of
Arrays	?	?
break	?	?
continue	?	?
Callback Function	?	?
Readability	Good	Excellent

Memory Tip

forEach() → Perform action on every array element

for...of → Iterate values with more control

In modern web development, forEach() is heavily used for:

- Product listings
- API data display
- Dynamic tables
- User cards
- Shopping carts

- Dashboard reports

5. map()

Creates a new array by transforming each element.

```
let nums = [1,2,3,4];

let square = nums.map(num => num*num);

console.log(square);
```

Output

[1,4,9,16]

Real Example

```
let prices = [100,200,300];

let gstPrice = prices.map(price => price + 18);

console.log(gstPrice);
```

Output

[118,218,318]

What is map()?

The map() method creates a **new array** by applying a function to every element of an existing array.

Syntax

```
let newArray = array.map(function(value)
{
   return modifiedValue;
});
```

---

1. Double Every Number

```
let numbers = [1,2,3,4,5];

let result = numbers.map(function(num)
{
   return num * 2;
});

console.log(result);
```

Output

```
[2,4,6,8,10]
```

---

2. Square Every Number

```
let numbers = [1,2,3,4,5];

let squares = numbers.map(num => num ** 2);

console.log(squares);
```

Output

```
[1,4,9,16,25]
```

3. Convert Celsius to Fahrenheit

Formula:

F=(C×95)+32F=(C\times\frac{9}{5})+32F=(C×59)+32

```
let celsius = [0,10,20,30,40];

let fahrenheit = celsius.map(temp =>
(temp * 9/5) + 32);

console.log(fahrenheit);
```

Output

```
[32,50,68,86,104]
```

---

4. Add GST (18%)

```
let prices = [1000,2000,3000];

let finalPrice = prices.map(price =>
price + (price * 18/100));

console.log(finalPrice);
```

Output

```
[1180,2360,3540]
```

---

5. Convert Names to Uppercase

```
let names =
["rahul","aman","vicky"];

let result = names.map(name =>
name.toUpperCase());

console.log(result);
```

Output

```
["RAHUL","AMAN","VICKY"]
```

---

6. Add ₹ Symbol to Prices

```
let prices =
[1000,2000,3000];

let result = prices.map(price =>
"₹" + price);

console.log(result);
```

Output

```
["₹1000","₹2000","₹3000"]
```

---

7. Extract Student Names

```
let students =
[
   {name:"Rahul", age:20},
   {name:"Aman", age:21},
```

```
    {name:"Vicky", age:25}
];

let names =
students.map(student =>
student.name);

console.log(names);
```

Output

```
["Rahul","Aman","Vicky"]
```

---

8. Generate Email IDs

```
let users =
["rahul","aman","vicky"];

let emails =
users.map(user =>
user + "@gmail.com");

console.log(emails);
```

Output

```
[
 "rahul@gmail.com",
 "aman@gmail.com",
 "vicky@gmail.com"
]
```

---

9. Convert Array of Strings to Numbers

```
let numbers =
["10","20","30","40"];

let result =
numbers.map(Number);

console.log(result);
```

Output

```
[10,20,30,40]
```

---

10. Generate Product Cards

```
let products =
[
   "Laptop",
   "Mobile",
   "Tablet"
];

let html =
products.map(product =>
`<h2>${product}</h2>`);

console.log(html);
```

Output

```
[
 "<h2>Laptop</h2>",
```

```
 "<h2>Mobile</h2>",
 "<h2>Tablet</h2>"
]
```

---

11. Employee Salary Increment

```
let salaries =
[30000,40000,50000];

let updated =
salaries.map(salary =>
salary + 5000);

console.log(updated);
```

Output

```
[35000,45000,55000]
```

---

12. Convert Names to Objects

```
let names =
["Rahul","Aman","Vicky"];

let users =
names.map(name =>
({
   username:name
}));

console.log(users);
```

Output

```
[
 {username:"Rahul"},
 {username:"Aman"},
 {username:"Vicky"}
]
```

---

13. Create Multiplication Table of 5

```
let numbers =
[1,2,3,4,5,6,7,8,9,10];

let table =
numbers.map(num =>
`5 x ${num} = ${5*num}`);

console.log(table);
```

---

14. Shopping Cart Discount

10% Discount

```
let prices =
[1000,2000,3000];

let discounted =
prices.map(price =>
price - (price*10/100));

console.log(discounted);
```

Output

[900,1800,2700]

---

15. Extract Product Prices

```
let products =
[
 {name:"Laptop", price:50000},
 {name:"Mobile", price:25000},
 {name:"Tablet", price:15000}
];

let prices =
products.map(product =>
product.price);

console.log(prices);
```

Output

[50000,25000,15000]

Real Web Development Examples

16. User Cards

```
let users =
[
   "Rahul",
   "Aman",
   "Vicky"
```

```
];

document.body.innerHTML =
users.map(user =>
`
<div class="card">
   <h2>${user}</h2>
</div>
`).join("");
```

---

17. Product Listing

```
let products =
[
   {name:"Laptop",price:50000},
   {name:"Mobile",price:25000}
];

let html =
products.map(product =>
`
<div>
<h2>${product.name}</h2>
<p>₹${product.price}</p>
</div>
`).join("");

document.body.innerHTML = html;
```

---

18. Student Report Card

```
let marks =
[80,90,70,85];

let grades =
marks.map(mark =>
{
   if(mark >= 90) return "A+";
   if(mark >= 80) return "A";
   if(mark >= 70) return "B";

   return "C";
});

console.log(grades);
```

Output

```
["A","A+","B","A"]
```

---

19. OTP Generator

```
let otp =
Array(6)
.fill(0)
.map(() =>
Math.floor(Math.random()*10))
.join("");

console.log(otp);
```

Output

483921

---

20. Generate IDs

```
let users =
["Rahul","Aman","Vicky"];

let ids =
users.map((user,index) =>
({
   id:index+1,
   name:user
}));

console.log(ids);
```

Output

```
[
 {id:1,name:"Rahul"},
 {id:2,name:"Aman"},
 {id:3,name:"Vicky"}
]
```

Interview Questions

Q1. Difference between forEach() and map()?

Feature	map()	forEach()
Returns New Array	?	?
Modifies Data	?	?

Feature	map()	forEach()
Chaining	⍰	⍰
Used in React	⍰ Very Common	Less Common

Q2. Why is map() heavily used in React?

```
let users =
["Rahul","Aman","Vicky"];

users.map(user =>
<h2>{user}</h2>);
```

map() is the most commonly used method for rendering dynamic lists in React applications.

Most Important Practice Questions

Basic

1. Double Numbers
2. Square Numbers
3. Cube Numbers
4. Add 100 to Every Element
5. Convert Names to Uppercase

Intermediate

6. Add GST
7. Apply Discount
8. Generate Emails
9. Convert String Array to Number Array

10. Extract Object Properties

Advanced

11. Product Cards
12. Student Grades
13. Employee Salary Increment
14. Generate OTP
15. Dynamic User Dashboard

Pro Level

16. E-commerce Product Listing
17. Movie Cards
18. Blog Cards
19. Student Report Card
20. React Component Rendering

Memory Trick

forEach() → Do Something

map() → Transform Something

Example:

```
let numbers = [1,2,3];

let result =
numbers.map(num => num * 2);

console.log(result);
```

Output:

[2,4,6]

map() always creates and returns a **new transformed array**, which is why it's one of the most important array methods in modern JavaScript.

---

6. filter()

Filters elements based on a condition.

let nums = [10,15,20,25,30];

let even = nums.filter(num => num%2==0);

console.log(even);

Output

[10,20,30]

Students Example

let marks = [35,20,80,50,25];

let pass = marks.filter(mark => mark >=33);

console.log(pass);

Output

[35,80,50]

What is map()?

The map() method creates a **new array** by applying a function to every element of an existing array.

Syntax

```
let newArray = array.map(function(value)
{
   return modifiedValue;
});
```

---

1. Double Every Number

```
let numbers = [1,2,3,4,5];

let result = numbers.map(function(num)
{
   return num * 2;
});

console.log(result);
```

Output

```
[2,4,6,8,10]
```

---

2. Square Every Number

```
let numbers = [1,2,3,4,5];
```

```
let squares = numbers.map(num => num ** 2);

console.log(squares);
```

Output

[1,4,9,16,25]

---

3. Convert Celsius to Fahrenheit

Formula:

F=(C×95)+32F=(C\times\frac{9}{5})+32F=(C×59)+32

```
let celsius = [0,10,20,30,40];

let fahrenheit = celsius.map(temp =>
(temp * 9/5) + 32);

console.log(fahrenheit);
```

Output

[32,50,68,86,104]

---

4. Add GST (18%)

```
let prices = [1000,2000,3000];

let finalPrice = prices.map(price =>
price + (price * 18/100));

console.log(finalPrice);
```

Output

[1180,2360,3540]

---

5. Convert Names to Uppercase

let names =
["rahul","aman","vicky"];

let result = names.map(name =>
name.toUpperCase());

console.log(result);

Output

["RAHUL","AMAN","VICKY"]

---

6. Add ₹ Symbol to Prices

let prices =
[1000,2000,3000];

let result = prices.map(price =>
"₹" + price);

console.log(result);

Output

["₹1000","₹2000","₹3000"]

---

7. Extract Student Names

```
let students =
[
   {name:"Rahul", age:20},
   {name:"Aman", age:21},
   {name:"Vicky", age:25}
];

let names =
students.map(student =>
student.name);

console.log(names);
```

Output

```
["Rahul","Aman","Vicky"]
```

---

8. Generate Email IDs

```
let users =
["rahul","aman","vicky"];

let emails =
users.map(user =>
user + "@gmail.com");

console.log(emails);
```

Output

```
[
 "rahul@gmail.com",
 "aman@gmail.com",
 "vicky@gmail.com"
]
```

---

9. Convert Array of Strings to Numbers

```
let numbers =
["10","20","30","40"];

let result =
numbers.map(Number);

console.log(result);
```

Output

```
[10,20,30,40]
```

---

10. Generate Product Cards

```
let products =
[
   "Laptop",
   "Mobile",
   "Tablet"
];

let html =
products.map(product =>
```

```
`<h2>${product}</h2>`);

console.log(html);
```

Output

```
[
 "<h2>Laptop</h2>",
 "<h2>Mobile</h2>",
 "<h2>Tablet</h2>"
]
```

---

11. Employee Salary Increment

```
let salaries =
[30000,40000,50000];

let updated =
salaries.map(salary =>
salary + 5000);

console.log(updated);
```

Output

```
[35000,45000,55000]
```

---

12. Convert Names to Objects

```
let names =
["Rahul","Aman","Vicky"];
```

```
let users =
names.map(name =>
({
   username:name
}));

console.log(users);
```

Output

```
[
 {username:"Rahul"},
 {username:"Aman"},
 {username:"Vicky"}
]
```

---

13. Create Multiplication Table of 5

```
let numbers =
[1,2,3,4,5,6,7,8,9,10];

let table =
numbers.map(num =>
`5 x ${num} = ${5*num}`);

console.log(table);
```

---

14. Shopping Cart Discount

10% Discount

```
let prices =
[1000,2000,3000];

let discounted =
prices.map(price =>
price - (price*10/100));

console.log(discounted);
```

Output

```
[900,1800,2700]
```

---

15. Extract Product Prices

```
let products =
[
 {name:"Laptop", price:50000},
 {name:"Mobile", price:25000},
 {name:"Tablet", price:15000}
];

let prices =
products.map(product =>
product.price);

console.log(prices);
```

Output

```
[50000,25000,15000]
```

Real Web Development Examples

16. User Cards

```
let users =
[
   "Rahul",
   "Aman",
   "Vicky"
];

document.body.innerHTML =
users.map(user =>
`
<div class="card">
   <h2>${user}</h2>
</div>
`).join("");
```

---

17. Product Listing

```
let products =
[
   {name:"Laptop",price:50000},
   {name:"Mobile",price:25000}
];

let html =
products.map(product =>
`
<div>
<h2>${product.name}</h2>
```

```
<p>₹${product.price}</p>
</div>
`).join("");

document.body.innerHTML = html;
```

---

18. Student Report Card

```
let marks =
[80,90,70,85];

let grades =
marks.map(mark =>
{
   if(mark >= 90) return "A+";
   if(mark >= 80) return "A";
   if(mark >= 70) return "B";

   return "C";
});

console.log(grades);
```

Output

```
["A","A+","B","A"]
```

---

19. OTP Generator

```
let otp =
Array(6)
```

```
.fill(0)
.map(() =>
Math.floor(Math.random()*10))
.join("");

console.log(otp);
```

Output

```
483921
```

---

20. Generate IDs

```
let users =
["Rahul","Aman","Vicky"];

let ids =
users.map((user,index) =>
({
   id:index+1,
   name:user
}));

console.log(ids);
```

Output

```
[
 {id:1,name:"Rahul"},
 {id:2,name:"Aman"},
 {id:3,name:"Vicky"}
]
```

Interview Questions

Q1. Difference between forEach() and map()?

Feature	map()	forEach()
Returns New Array	?	?
Modifies Data	?	?
Chaining	?	?
Used in React	? Very Common	Less Common

Q2. Why is map() heavily used in React?

```
let users =
["Rahul","Aman","Vicky"];

users.map(user =>
<h2>{user}</h2>);
```

map() is the most commonly used method for rendering dynamic lists in React applications.

Most Important Practice Questions

Basic

1. Double Numbers
2. Square Numbers
3. Cube Numbers
4. Add 100 to Every Element

5. Convert Names to Uppercase

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Pro Level

16. E-commerce Product Listing
17. Movie Cards
18. Blog Cards
19. Student Report Card
20. React Component Rendering

Memory Trick

forEach() → Do Something

map() → Transform Something

Example:

```
let numbers = [1,2,3];

let result =
numbers.map(num => num * 2);

console.log(result);
```

Output:

```
[2,4,6]
```

map() always creates and returns a **new transformed array**, which is why it's one of the most important array methods in modern JavaScript.

---

7. reduce()

Used to calculate a single value.

Sum of Array

```
let nums = [10,20,30,40];

let sum = nums.reduce(
   (total,num) => total + num
);

console.log(sum);
```

Output

```
100
```

Maximum Number

```
let nums = [10,80,40,20];

let max = nums.reduce(
   (a,b) => a>b ? a : b
);

console.log(max);
```

Output

```
80
```

JavaScript filter() Method - Interesting & Useful Questions with Answers

What is filter()?

filter() creates a **new array** containing only those elements that satisfy a condition.

Syntax

```
let result = array.filter(function(value)
{
   return condition;
});
```

---

1. Filter Even Numbers

```
let numbers = [1,2,3,4,5,6,7,8,9,10];

let evenNumbers = numbers.filter(num => num % 2 === 0);

console.log(evenNumbers);
```

Output

[2,4,6,8,10]

---

2. Filter Odd Numbers

```
let numbers = [1,2,3,4,5,6,7,8,9,10];

let oddNumbers = numbers.filter(num => num % 2 !== 0);

console.log(oddNumbers);
```

Output

[1,3,5,7,9]

---

3. Filter Positive Numbers

```
let numbers = [-10,20,-5,30,40,-8];

let positive = numbers.filter(num => num > 0);

console.log(positive);
```

Output

[20,30,40]

---

4. Filter Numbers Greater Than 50

```
let numbers = [20,40,60,80,100];

let result = numbers.filter(num => num > 50);

console.log(result);
```

Output

```
[60,80,100]
```

---

5. Filter Voters (Age ≥ 18)

```
let ages = [12,18,25,15,30,10];

let voters = ages.filter(age => age >= 18);

console.log(voters);
```

Output

```
[18,25,30]
```

---

6. Filter Long Names

```
let names =
["Rahul","Aman","Vicky","Al","Riya"];

let result =
names.filter(name =>
```

name.length > 4);

console.log(result);

Output

["Rahul","Vicky"]

---

7. Filter Words Starting with "A"

let names =
["Aman","Rahul","Ankit","Vicky"];

let result =
names.filter(name =>
name.startsWith("A"));

console.log(result);

Output

["Aman","Ankit"]

---

8. Filter Expensive Products

let prices =
[500,2000,10000,800,15000];

let expensive =
prices.filter(price =>
price > 5000);

console.log(expensive);

Output

[10000,15000]

---

9. Filter Passed Students

let marks =
[20,40,80,30,90];

let passed =
marks.filter(mark =>
mark >= 33);

console.log(passed);

Output

[40,80,90]

---

10. Filter Failed Students

let marks =
[20,40,80,30,90];

let failed =
marks.filter(mark =>
mark < 33);

console.log(failed);

Output

[20,30]

---

11. Filter Adult Users (Objects)

```
let users =
[
   {name:"Rahul", age:15},
   {name:"Aman", age:20},
   {name:"Vicky", age:25}
];

let adults =
users.filter(user =>
user.age >= 18);

console.log(adults);
```

Output

```
[
 {name:"Aman", age:20},
 {name:"Vicky", age:25}
]
```

---

12. Filter Available Products

```
let products =
[
 {name:"Laptop", stock:true},
 {name:"Mobile", stock:false},
```

```
 {name:"Tablet", stock:true}
];

let available =
products.filter(product =>
product.stock);

console.log(available);
```

Output

```
[
 {name:"Laptop", stock:true},
 {name:"Tablet", stock:true}
]
```

---

13. Filter Gmail Users

```
let emails =
[
"abc@gmail.com",
"xyz@yahoo.com",
"test@gmail.com"
];

let gmailUsers =
emails.filter(email =>
email.includes("@gmail.com"));

console.log(gmailUsers);
```

Output

```
[
"abc@gmail.com",
"test@gmail.com"
]
```

---

14. Filter Students with A Grade

```
let students =
[
 {name:"Rahul", grade:"A"},
 {name:"Aman", grade:"B"},
 {name:"Vicky", grade:"A"}
];

let toppers =
students.filter(student =>
student.grade === "A");

console.log(toppers);
```

---

15. Filter Prime Numbers

```
let numbers =
[2,3,4,5,6,7,8,9,11];

let primes =
numbers.filter(num =>
{
   if(num < 2) return false;

   for(let i=2; i<num; i++)
```

```
    {
        if(num % i === 0)
        {
            return false;
        }
    }

    return true;
});

console.log(primes);
```

Output

```
[2,3,5,7,11]
```

Real Web Development Examples

16. Search Products

```
let products =
[
"Laptop",
"Mobile",
"Tablet",
"Mouse"
];

let search = "Mo";

let result =
products.filter(product =>
```

```
product.includes(search));

console.log(result);
```

Output

["Mobile","Mouse"]

---

17. Filter Active Users

```
let users =
[
 {name:"Rahul", active:true},
 {name:"Aman", active:false},
 {name:"Vicky", active:true}
];

let activeUsers =
users.filter(user =>
user.active);

console.log(activeUsers);
```

---

18. Filter High Salary Employees

```
let employees =
[
 {name:"Rahul", salary:30000},
 {name:"Aman", salary:80000},
 {name:"Vicky", salary:50000}
];
```

```
let highSalary =
employees.filter(emp =>
emp.salary > 50000);

console.log(highSalary);
```

---

19. Filter Discounted Products

```
let products =
[
 {name:"Laptop", discount:true},
 {name:"Mobile", discount:false},
 {name:"Tablet", discount:true}
];

let offers =
products.filter(product =>
product.discount);

console.log(offers);
```

---

20. Filter In-Stock Products

```
let products =
[
 {name:"Laptop", quantity:5},
 {name:"Mobile", quantity:0},
 {name:"Tablet", quantity:10}
];
```

```
let stockProducts =
products.filter(product =>
product.quantity > 0);

console.log(stockProducts);
```

Output

```
[
 {name:"Laptop", quantity:5},
 {name:"Tablet", quantity:10}
]
```

Bonus Interview Questions

Q1. Difference Between map() and filter()

map()

Transforms data.

```
let nums = [1,2,3];

let result =
nums.map(num => num * 2);

console.log(result);
```

Output:

[2,4,6]

---

filter()

Selects data.

```
let nums = [1,2,3,4];

let result =
nums.filter(num => num % 2 === 0);

console.log(result);
```

Output:

```
[2,4]
```

Most Important Practice Questions

Basic

1. Filter Even Numbers
2. Filter Odd Numbers
3. Filter Positive Numbers
4. Filter Negative Numbers
5. Filter Numbers Greater Than 100

Intermediate

6. Filter Passed Students
7. Filter Adult Users
8. Filter Gmail Users
9. Filter Long Names
10. Filter Available Products

Advanced

11. Filter Prime Numbers

12. Filter High Salary Employees
13. Filter Active Users
14. Product Search System
15. Shopping Website Product Filters

Pro Level

16. E-commerce Search
17. Job Portal Filter
18. Student Result Filter
19. Employee Management Filter
20. Online Store Product Filter

Memory Trick

map() → Transform Data
filter() → Select Data
forEach()→ Perform Action

Example:

```
let numbers = [1,2,3,4,5];

let result =
numbers.filter(num => num > 3);

console.log(result);
```

Output:

[4,5]

filter() is one of the most commonly used methods in real-world applications for **searching, filtering products, filtering users,**

dashboards, e-commerce sites, admin panels, and API data processing.

---

8. find()

Returns first matching value.

```
let nums = [5,10,15,20];

let result = nums.find(num => num > 10);

console.log(result);
```

Output

```
15
```

JavaScript find() Method - Practical Questions with Answers

What is find()?

find() returns the **first element** that satisfies a condition.

Syntax

```
array.find(function(value)
{
   return condition;
});
```

Example

```
let numbers = [10,20,30,40];

let result = numbers.find(num => num > 25);

console.log(result);
```

Output

30

find() returns only the **first matching value**.

---

1. Find First Even Number

```
let numbers = [11,15,18,20,25];

let result = numbers.find(num => num % 2 === 0);

console.log(result);
```

Output

18

---

2. Find First Odd Number

```
let numbers = [2,4,6,7,8];

let result = numbers.find(num => num % 2 !== 0);

console.log(result);
```

Output

7

---

3. Find First Number Greater Than 100

```
let numbers = [20,50,80,120,150];

let result = numbers.find(num => num > 100);

console.log(result);
```

Output

120

---

4. Find First Negative Number

```
let numbers = [10,20,-5,-10];

let result = numbers.find(num => num < 0);

console.log(result);
```

Output

-5

---

5. Find First Voter

```
let ages = [12,15,17,18,22];
```

```
let voter = ages.find(age => age >= 18);

console.log(voter);
```

Output

18

---

6. Find Student with Marks Above 90

```
let marks = [70,80,95,85];

let topper = marks.find(mark => mark > 90);

console.log(topper);
```

Output

95

---

7. Find User by Name

```
let users =
[
   {id:1,name:"Rahul"},
   {id:2,name:"Aman"},
   {id:3,name:"Vicky"}
];

let user =
users.find(user =>
user.name === "Aman");
```

```
console.log(user);
```

Output

```
{
 id:2,
 name:"Aman"
}
```

---

8. Find Product by ID

```
let products =
[
 {id:101,name:"Laptop"},
 {id:102,name:"Mobile"},
 {id:103,name:"Tablet"}
];

let product =
products.find(product =>
product.id === 102);

console.log(product);
```

Output

```
{
 id:102,
 name:"Mobile"
}
```

---

9. Find First Expensive Product

```
let products =
[
 {name:"Mouse",price:500},
 {name:"Keyboard",price:1000},
 {name:"Laptop",price:50000}
];

let expensive =
products.find(product =>
product.price > 10000);

console.log(expensive);
```

Output

```
{
 name:"Laptop",
 price:50000
}
```

---

10. Find First Active User

```
let users =
[
 {name:"Rahul",active:false},
 {name:"Aman",active:false},
 {name:"Vicky",active:true}
];

let activeUser =
```

```
users.find(user =>
user.active);

console.log(activeUser);
```

Output

```
{
 name:"Vicky",
 active:true
}
```

---

11. Login System

```
let users =
[
{
username:"admin",
password:"1234"
},
{
username:"vicky",
password:"5678"
}
];

let login =
users.find(user =>
user.username === "admin"
&&
user.password === "1234"
);
```

```
console.log(login);
```

Output

```
{
 username:"admin",
 password:"1234"
}
```

---

12. Find First Available Product

```
let products =
[
{name:"Laptop",stock:0},
{name:"Mobile",stock:0},
{name:"Tablet",stock:5}
];

let available =
products.find(product =>
product.stock > 0);

console.log(available);
```

Output

```
{
 name:"Tablet",
 stock:5
}
```

---

13. Find First Failed Student

```
let marks =
[80,90,25,70,50];

let failed =
marks.find(mark =>
mark < 33);

console.log(failed);
```

Output

25

---

14. Find First Gmail User

```
let emails =
[
"abc@yahoo.com",
"xyz@gmail.com",
"test@gmail.com"
];

let gmail =
emails.find(email =>
email.includes("@gmail.com"));

console.log(gmail);
```

Output

xyz@gmail.com

---

15. Find First Premium Customer

```
let customers =
[
{name:"Rahul",premium:false},
{name:"Aman",premium:true},
{name:"Vicky",premium:true}
];

let premium =
customers.find(customer =>
customer.premium);

console.log(premium);
```

Output

```
{
 name:"Aman",
 premium:true
}
```

Real Web Development Examples

16. Search Product

```
let products =
[
{name:"Laptop"},
{name:"Mobile"},
{name:"Tablet"}
```

```
];

let result =
products.find(product =>
product.name === "Mobile");

console.log(result);
```

---

17. Find Employee by ID

```
let employees =
[
{id:1,name:"Rahul"},
{id:2,name:"Aman"},
{id:3,name:"Vicky"}
];

let emp =
employees.find(emp =>
emp.id === 3);

console.log(emp);
```

---

18. Find Highest Priority Task

```
let tasks =
[
{name:"Task1",priority:1},
{name:"Task2",priority:5},
{name:"Task3",priority:3}
];
```

```
let task =
tasks.find(task =>
task.priority === 5);

console.log(task);
```

---

19. Find Bank Account

```
let accounts =
[
{
accNo:101,
balance:5000
},
{
accNo:102,
balance:10000
}
];

let account =
accounts.find(acc =>
acc.accNo === 102);

console.log(account);
```

---

20. Find Student Record

```
let students =
[
```

```
{id:1,name:"Rahul"},
{id:2,name:"Aman"},
{id:3,name:"Vicky"}
];

let student =
students.find(student =>
student.id === 2);

console.log(student);
```

Output

```
{
 id:2,
 name:"Aman"
}
```

Most Important Interview Questions

Basic

1. Find First Even Number
2. Find First Odd Number
3. Find First Negative Number
4. Find First Number > 100
5. Find First Failed Student

Intermediate

6. Find User by Name
7. Find Product by ID
8. Find Employee by ID

9. Find Gmail User
10. Find Premium Customer

Advanced

11. Login System
12. Bank Account Search
13. Product Search
14. Student Record Search
15. Task Management System

Difference Between find() and filter()

find()

Returns first matching element.

```
let arr = [10,20,30,40];

console.log(
arr.find(num => num > 20)
);
```

Output:

30

---

filter()

Returns all matching elements.

```
let arr = [10,20,30,40];
```

```
console.log(
arr.filter(num => num > 20)
);
```

Output:

[30,40]

Memory Trick

find()   → First Match
filter() → All Matches

Example

```
let users =
[
{name:"Rahul"},
{name:"Aman"},
{name:"Vicky"}
];

let user =
users.find(
user => user.name === "Aman"
);

console.log(user);
```

Output:

```
{
 name:"Aman"
}
```

find() is heavily used in:

- Login systems
- Product search
- User search
- Student records
- Banking applications
- Admin dashboards
- E-commerce websites
- React applications

---

9. some()

Checks whether at least one element satisfies a condition.

```
let marks = [20,30,40,50];

let result = marks.some(mark => mark >= 40);

console.log(result);
```

Output

True

JavaScript find() Method - Practical Questions with Answers

What is find()?

find() returns the **first element** that satisfies a condition.

Syntax

```
array.find(function(value)
{
   return condition;
});
```

Example

```
let numbers = [10,20,30,40];

let result = numbers.find(num => num > 25);

console.log(result);
```

Output

30

find() returns only the **first matching value**.

---

1. Find First Even Number

```
let numbers = [11,15,18,20,25];

let result = numbers.find(num => num % 2 === 0);

console.log(result);
```

Output

18

---

2. Find First Odd Number

```
let numbers = [2,4,6,7,8];

let result = numbers.find(num => num % 2 !== 0);

console.log(result);
```

Output

7

---

3. Find First Number Greater Than 100

```
let numbers = [20,50,80,120,150];

let result = numbers.find(num => num > 100);

console.log(result);
```

Output

120

---

4. Find First Negative Number

```
let numbers = [10,20,-5,-10];

let result = numbers.find(num => num < 0);
```

```
console.log(result);
```

Output

```
-5
```

---

5. Find First Voter

```
let ages = [12,15,17,18,22];

let voter = ages.find(age => age >= 18);

console.log(voter);
```

Output

```
18
```

---

6. Find Student with Marks Above 90

```
let marks = [70,80,95,85];

let topper = marks.find(mark => mark > 90);

console.log(topper);
```

Output

```
95
```

---

7. Find User by Name

```
let users =
[
   {id:1,name:"Rahul"},
   {id:2,name:"Aman"},
   {id:3,name:"Vicky"}
];

let user =
users.find(user =>
user.name === "Aman");

console.log(user);
```

Output

```
{
 id:2,
 name:"Aman"
}
```

---

8. Find Product by ID

```
let products =
[
 {id:101,name:"Laptop"},
 {id:102,name:"Mobile"},
 {id:103,name:"Tablet"}
];

let product =
products.find(product =>
product.id === 102);
```

```
console.log(product);
```

Output

```
{
 id:102,
 name:"Mobile"
}
```

---

9. Find First Expensive Product

```
let products =
[
 {name:"Mouse",price:500},
 {name:"Keyboard",price:1000},
 {name:"Laptop",price:50000}
];

let expensive =
products.find(product =>
product.price > 10000);

console.log(expensive);
```

Output

```
{
 name:"Laptop",
 price:50000
}
```

---

10. Find First Active User

```
let users =
[
 {name:"Rahul",active:false},
 {name:"Aman",active:false},
 {name:"Vicky",active:true}
];

let activeUser =
users.find(user =>
user.active);

console.log(activeUser);
```

Output

```
{
 name:"Vicky",
 active:true
}
```

---

11. Login System

```
let users =
[
{
username:"admin",
password:"1234"
},
{
username:"vicky",
```

```
password:"5678"
}
];

let login =
users.find(user =>
user.username === "admin"
&&
user.password === "1234"
);

console.log(login);
```

Output

```
{
 username:"admin",
 password:"1234"
}
```

---

12. Find First Available Product

```
let products =
[
{name:"Laptop",stock:0},
{name:"Mobile",stock:0},
{name:"Tablet",stock:5}
];

let available =
products.find(product =>
product.stock > 0);
```

```
console.log(available);
```

Output

```
{
 name:"Tablet",
 stock:5
}
```

---

13. Find First Failed Student

```
let marks =
[80,90,25,70,50];

let failed =
marks.find(mark =>
mark < 33);

console.log(failed);
```

Output

```
25
```

---

14. Find First Gmail User

```
let emails =
[
"abc@yahoo.com",
"xyz@gmail.com",
"test@gmail.com"
```

```
];

let gmail =
emails.find(email =>
email.includes("@gmail.com"));

console.log(gmail);
```

Output

```
xyz@gmail.com
```

---

15. Find First Premium Customer

```
let customers =
[
{name:"Rahul",premium:false},
{name:"Aman",premium:true},
{name:"Vicky",premium:true}
];

let premium =
customers.find(customer =>
customer.premium);

console.log(premium);
```

Output

```
{
 name:"Aman",
```

```
 premium:true
}
```

Real Web Development Examples

16. Search Product

```
let products =
[
{name:"Laptop"},
{name:"Mobile"},
{name:"Tablet"}
];

let result =
products.find(product =>
product.name === "Mobile");

console.log(result);
```

---

17. Find Employee by ID

```
let employees =
[
{id:1,name:"Rahul"},
{id:2,name:"Aman"},
{id:3,name:"Vicky"}
];

let emp =
employees.find(emp =>
```

```
emp.id === 3);

console.log(emp);
```

---

18. Find Highest Priority Task

```
let tasks =
[
{name:"Task1",priority:1},
{name:"Task2",priority:5},
{name:"Task3",priority:3}
];

let task =
tasks.find(task =>
task.priority === 5);

console.log(task);
```

---

19. Find Bank Account

```
let accounts =
[
{
accNo:101,
balance:5000
},
{
accNo:102,
balance:10000
}
```

```
];

let account =
accounts.find(acc =>
acc.accNo === 102);

console.log(account);
```

---

20. Find Student Record

```
let students =
[
{id:1,name:"Rahul"},
{id:2,name:"Aman"},
{id:3,name:"Vicky"}
];

let student =
students.find(student =>
student.id === 2);

console.log(student);
```

Output

```
{
 id:2,
 name:"Aman"
}
```

Most Important Interview Questions

Basic

1. Find First Even Number
2. Find First Odd Number
3. Find First Negative Number
4. Find First Number > 100
5. Find First Failed Student

Intermediate

6. Find User by Name
7. Find Product by ID
8. Find Employee by ID
9. Find Gmail User
10. Find Premium Customer

Advanced

11. Login System
12. Bank Account Search
13. Product Search
14. Student Record Search
15. Task Management System

Difference Between find() and filter()

find()

Returns first matching element.

```
let arr = [10,20,30,40];

console.log(
```

```
arr.find(num => num > 20)
);
```

Output:

30

---

filter()

Returns all matching elements.

```
let arr = [10,20,30,40];

console.log(
arr.filter(num => num > 20)
);
```

Output:

[30,40]

Memory Trick

find()   → First Match
filter() → All Matches

Example

```
let users =
[
{name:"Rahul"},
{name:"Aman"},
{name:"Vicky"}
```

```
];

let user =
users.find(
user => user.name === "Aman"
);

console.log(user);
```

Output:

```
{
 name:"Aman"
}
```

find() is heavily used in:

- Login systems
- Product search
- User search
- Student records
- Banking applications
- Admin dashboards
- E-commerce websites
- React applications

---

10. every()

Checks whether all elements satisfy a condition.

```
let marks = [40,50,60,70];

let result = marks.every(mark => mark >= 33);

console.log(result);
```

Output

True

JavaScript every() Method - Practical & Real-World Questions with Answers

What is every()?

The every() method checks whether **all elements** satisfy a condition.

- Returns true → if all elements pass
- Returns false → if even one element fails

```
let result = array.every(condition);
```

---

1. Check All Students Passed

```
let marks = [80, 75, 90, 65];

let result = marks.every(mark => mark >= 33);

console.log(result);
```

Output

true

---

2. Check All Numbers Are Positive

```
let numbers = [10, 20, 30, 40];

let result = numbers.every(num => num > 0);

console.log(result);
```

Output

```
true
```

---

3. Check All Numbers Are Even

```
let numbers = [2, 4, 6, 8, 10];

let result = numbers.every(num => num % 2 === 0);

console.log(result);
```

Output

```
true
```

---

4. Check All Employees Are Adults

```
let ages = [22, 35, 40, 28];

let result = ages.every(age => age >= 18);

console.log(result);
```

Output

true

---

5. Check All Products Are In Stock

```
let products = [
   {name:"Laptop", stock:true},
   {name:"Mobile", stock:true},
   {name:"Tablet", stock:true}
];

let result = products.every(product => product.stock);

console.log(result);
```

Output

true

---

6. Check All Students Have A Grade

```
let students = [
   {name:"Rahul", grade:"A"},
   {name:"Aman", grade:"A"},
   {name:"Vicky", grade:"A"}
];

let result =
students.every(student => student.grade === "A");

console.log(result);
```

Output

true

---

7. Password Validation System

Minimum 8 Characters

```
let passwords =
["admin123","welcome12","hello1234"];

let result =
passwords.every(pass =>
pass.length >= 8);

console.log(result);
```

Output

true

---

8. Check All Email Addresses Are Gmail

```
let emails =
[
"rahul@gmail.com",
"aman@gmail.com",
"vicky@gmail.com"
];

let result =
emails.every(email =>
```

```
email.endsWith("@gmail.com"));

console.log(result);
```

Output

```
true
```

---

9. Check All Marks Above 50

```
let marks =
[60,70,80,90];

let result =
marks.every(mark =>
mark > 50);

console.log(result);
```

Output

```
true
```

---

10. Check All Numbers Are Integers

```
let numbers =
[10,20,30,40];

let result =
numbers.every(Number.isInteger);

console.log(result);
```

Output

true

Real-World Practical Questions

11. E-Commerce Cart Validation

Check all products are available before checkout.

```
let cart =
[
{name:"Laptop", stock:true},
{name:"Mouse", stock:true},
{name:"Keyboard", stock:true}
];

let canCheckout =
cart.every(item =>
item.stock);

console.log(canCheckout);
```

Output

true

---

12. Scholarship Eligibility

All marks must be above 75.

```
let marks =
[80,85,90,78];
```

```
let eligible =
marks.every(mark =>
mark >= 75);

console.log(eligible);
```

Output

```
true
```

---

13. Online Exam Verification

All questions attempted?

```
let answers =
[
true,
true,
true,
true,
true
];

let submitted =
answers.every(answer =>
answer === true);

console.log(submitted);
```

Output

```
true
```

---

14. Loan Approval System

All documents uploaded?

```
let documents =
[
true,
true,
true,
true
];

let approved =
documents.every(doc =>
doc);

console.log(approved);
```

Output

```
true
```

---

15. Attendance System

Check all students are present.

```
let attendance =
[
true,
true,
true,
```

```
true
];

let result =
attendance.every(status =>
status);

console.log(result);
```

Output

true

Advanced Questions

16. Check All Salaries Above ₹30,000

```
let employees =
[
{salary:40000},
{salary:50000},
{salary:60000}
];

let result =
employees.every(emp =>
emp.salary > 30000);

console.log(result);
```

Output

true

17. Check All Usernames Are Unique Length > 4

```
let users =
["rahul","aman123","vicky"];

let result =
users.every(user =>
user.length > 4);

console.log(result);
```

Output

```
true
```

---

18. Check All Products Have Price

```
let products =
[
{name:"Laptop",price:50000},
{name:"Mobile",price:25000}
];

let result =
products.every(product =>
product.price > 0);

console.log(result);
```

Output

true

---

19. Validate Registration Form

```
let formData =
{
name:"Vicky",
email:"vicky@gmail.com",
phone:"9876543210"
};

let result =
Object.values(formData)
.every(value =>
value !== "");

console.log(result);
```

Output

true

---

20. Check All Subjects Passed

```
let reportCard =
{
html:80,
css:70,
js:90,
react:65
};
```

```
let result =
Object.values(reportCard)
.every(mark =>
mark >= 33);

console.log(result);
```

Output

true

Interview Question: Difference Between some() and every()

some()

At least one element must satisfy.

```
let numbers = [10,20,30,40];

console.log(
numbers.some(num => num > 35)
);
```

Output:

true

---

every()

All elements must satisfy.

```
let numbers = [10,20,30,40];

console.log(
numbers.every(num => num > 35)
);
```

Output:

false

Most Important Practice Questions

Basic

1. Check All Numbers Positive
2. Check All Numbers Even
3. Check All Students Passed
4. Check All Ages Adult
5. Check All Emails Valid

Intermediate

6. Scholarship Eligibility
7. Product Availability Check
8. Employee Salary Validation
9. Password Length Validation
10. Attendance System

Advanced

11. Registration Form Validation
12. Loan Approval System
13. Online Exam Submission
14. E-Commerce Checkout Validation

15. Student Report Card Validation

Pro-Level Projects

16. Banking KYC Verification
17. HR Employee Validation
18. Online Shopping Checkout
19. School Management System
20. Hospital Patient Registration Validation

Memory Trick

find() → First Match

filter() → All Matches

some() → At Least One Match

every() → All Must Match

Quick Example

```
let marks = [80,90,75,88];

let result =
marks.every(mark => mark >= 33);

console.log(result);
```

Output

true

Because **every student passed the exam**.